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Investor Presentation | Asia Pacific

Global/China AI Semis: TSMC/MediaTek Preview, China CoWoS/Memory, CPO/NPO

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GREATER CHINA TECHNOLOGY SEMICONDUCTORS	
Asia Pacific	
Industry View	Attractive

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Strong AI Semi Outlook

- Top ideas:
 - **OW:**
 - **AI:** MediaTek (Top Pick), TSMC, SMIC, Aspeed, Alchip, KYEC, ASE, ASMPT, AllRing, GUC
 - **Memory:** CXMT, Macronix, AP Memory (Top Pick), Nanya Tech, Winbond, GigaDevice
 - **China – AI/Semis/WFE:** Iluvatar, Cambricon, Hygon, NAURA Tech, AMEC
 - **Testing Equipment and Consumables:** MPI, Hon Precision
 - **Mature nodes:** UMC
 - **EW:** OmniVision, Phison, MetaX, Realtek, GlobalWafers, FOCl, WIN Semi
 - **UW:** Silergy, ASMedia

Long-term demand drivers:

- **Chip inflation:** We expect “price elasticity” to affect demand for tech products. Rising wafer, OSAT, and memory costs create more margin headwinds for chip designers in 2026.
- **AI cannibalization:** Besides demand weakness (AI replacing some human jobs), we see the semi supply chain also prioritizing AI semis over non-AI semis, e.g., T-Glass and memory shortages.
- **China – AI: DeepSeek is triggering inferencing AI demand:** DeepSeek has demonstrated cheaper inferencing. The local foundry supply chain is also becoming more capable in terms of AI GPU production.

Valuation Comparison: Foundry, Back-end, Memory, IDM and Semi Cap

								Dividend	FCF	Market	3M Avg. Daily	P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)		
Ticker	Company	Price	Curr	Price	Upside/		Rating	Yield	Yield	Cap	Trading	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
10/8/2026			ency	Target	Downside/			(%)	(%)	(US\$M)	(US\$M)												
Foundry																							
2330.TW	TSMC	2,550.0	TWD	3,088.0	21%	O		1.4%	1.8%	2,087,731	2,117.7	22.1	16.7	13.4	74%	32%	24%	45%	42%	38%	8.4	6.0	4.4
2303.TW	UMC	147.5	TWD	188.0	27%	O		5.1%	5.3%	58,798	728.5	21.0	20.8	16.3	111%	1%	27%	21%	20%	23%	4.2	4.0	3.7
0981.HK	SMIC	56.8	HKD	85.0	50%	O		0.0%	-5.0%	51,016	702.5	NM	NM	NM	126%	65%	18%	7%	11%	11%	NM	NM	NM
5347.TWO	Vanguard	186.5	TWD	180.0	-3%	E		4.9%	3.1%	10,944	102.2	34.2	27.6	22.4	31%	24%	23%	12%	14%	17%	4.0	3.8	3.6
1347.HK	Hua Hong	95.6	HKD	128.0	34%	E		0.1%	-4.6%	21,242	524.0	NM	NM	NM	491%	76%	30%	4%	7%	8%	NM	NM	NM
6770.TW	PSMC	72.2	TWD	111.0	54%	O		0.0%	5.6%	9,630	368.5	NM	NM	NM	-151%	301%	28%	4%	11%	12%	1.5	1.2	1.0
3105.TWO	WIN Semi	581.0	TWD	600.0	3%	E		0.5%	-3.8%	2,264	330.9	NM	51.5	28.6	117%	31%	80%	9%	11%	18%	5.5	5.4	5.0
8086.TWO	AWSC	154.0	TWD	130.0	-16%	U		1.1%	-3.9%	956	15.9	24.3	21.6	17.4	89%	13%	24%	14%	14%	16%	3.3	2.9	2.5
Mean:		1.6%						-0.2%				29.2	33.5	22.8	111%	68%	32%	15%	16%	18%	4.5	3.9	3.4
Median:		0.8%						-1.0%				23.2	21.6	17.4	100%	32%	26%	10%	13%	16%	4.1	3.9	3.6
Memory																							
688825.SS	CXMT	50.5	CNY	88.0	74%	O		NA	NA	469,405	3,421.9	16.8	10.6	8.0	NM	59%	33%	105%	69%	51%	11.1	5.4	3.2
603986.SS	Giga Device	334.4	CNY	750.0	124%	O		0.3%	4.1%	31,026	3,003.4	9.6	7.4	6.4	1302%	30%	15%	78%	57%	45%	5.2	3.3	2.4
2408.TW	Nanya Tech	532.0	TWD	580.0	9%	O		0.0%	13.6%	51,775	1,110.3	8.6	4.7	7.2	NM	NM	NM	77%	68%	29%	4.4	2.2	1.6
2344.TW	Winbond	177.0	TWD	288.0	63%	O		0.0%	16.6%	25,151	669.0	7.1	3.6	3.0	NM	NM	NM	68%	67%	46%	3.6	1.8	1.1
8299.TWO	Phison	2,060.0	TWD	2,588.0	26%	E		1.5%	1.6%	14,178	265.6	5.5	15.6	36.5	839%	-65%	-57%	83%	22%	10%	3.3	3.6	3.7
SIMO.O	Silicon Motion	261.9	USD	400.0	53%	O		2.2%	2.1%	8,979	247.6	22.8	12.6	9.5	263%	81%	33%	38%	47%	45%	7.3	5.1	3.7
2337.TW	Macronix	116.0	TWD	220.0	90%	O		4.6%	17.2%	6,796	207.6	NM	NM	NM	-1421%	159%	5%	69%	81%	46%	2.4	1.0	0.6
Mean:		1.4%						9.2%				11.7	9.1	11.8	246%	53%	6%	74%	59%	39%	5.3	3.2	2.3
Median:		0.9%						8.8%				9.1	9.0	7.6	551%	59%	15%	77%	67%	45%	4.4	3.3	2.4
Backend																							
3711.TW	ASEH	744.0	TWD	858.0	15%	O		2.2%	6.0%	102,505	383.1	40.4	22.6	13.0	100%	79%	74%	20%	30%	41%	7.5	6.2	4.7
600584.SS	JCET	62.6	CNY	90.2	44%	O		0.4%	5.7%	15,562	1,531.2	42.6	30.5	23.4	68%	40%	30%	8%	11%	13%	3.5	3.2	2.8
688820.SS	SJ Semi	106.0	CNY	118.4	12%	E		NA	NA	27,418	316.9	179.3	122.0	89.4	3%	47%	37%	6%	8%	9%	9.6	8.9	8.1
601231.SS	USI	23.1	CNY	39.3	70%	O		1.4%	-7.4%	7,657	120.9	22.5	15.2	12.0	23%	48%	27%	11%	15%	17%	2.5	2.2	1.9
2449.TW	KYEC	304.5	TWD	348.0	14%	O		1.9%	1.7%	12,345	205.2	33.6	22.3	16.1	1%	51%	38%	21%	28%	32%	6.9	5.7	4.7
3363.TWO	FOCI	667.0	TWD	688.0	3%	E		0.2%	4.8%	2,215	72.4	NM	NM	NM	NM	NM	NM	-2%	39%	75%	NM	NM	NM
Mean:		1.5%						1.5%				28.1	18.8	14.0	39%	53%	41%	14%	18%	23%	6.0	5.2	4.4
Median:		1.7%						3.7%				28.1	18.8	14.0	23%	48%	37%	11%	15%	17%	6.9	5.7	4.7
Power Semis/ IDM																							
688396.SS	CR Micro	52.0	CNY	51.6	-1%	U		0.2%	1.6%	9,565	203.4	41.2	28.9	21.2	153%	43%	37%	7%	9%	11%	2.6	2.4	2.2
600460.SS	Silan Micro	29.6	CNY	26.9	-9%	U		0.5%	2.1%	6,852	485.7	38.2	24.3	18.8	224%	57%	29%	10%	14%	16%	3.5	3.1	2.7
603290.SS	StarPower	78.9	CNY	110.0	40%	E		1.0%	0.0%	2,623	100.3	58.8	33.2	21.4	-21%	77%	56%	5%	8%	11%	2.6	2.4	2.2
300373.SZ	Yangjie	81.0	CNY	136.0	68%	O		1.4%	-2.8%	6,076	234.0	27.0	19.1	14.6	29%	41%	31%	15%	19%	21%	3.9	3.3	2.8
Mean:		0.8%						0.2%				41.3	26.4	19.0	96%	55%	38%	9%	12%	14%	3.2	2.8	2.5
Median:		0.7%						0.8%				39.7	26.6	20.0	91%	50%	34%	8%	11%	13%	3.1	2.8	2.5
Semicap / Equipment/ Materials/ Supply Chain																							
002371.SZ	NAURA	600.3	CNY	788.0	31%	O		0.6%	-0.9%	60,432	1,034.5	54.9	33.9	27.0	24%	62%	26%	17%	23%	24%	8.7	7.2	6.0
688012.SS	AMEC	313.2	CNY	428.0	37%	O		0.0%	17.6%	41,537	1,214.7	52.4	45.6	34.7	86%	15%	32%	21%	21%	22%	10.6	8.6	6.8
7769.TW	Hon Precision	5,925.0	TWD	10,008.0	69%	O		2.5%	NA	33,658	153.4	45.9	26.6	15.9	69%	72%	67%	37%	53%	65%	NM	12.4	8.9
6223.TWO	MPI	5,575.0	TWD	8,000.0	43%	O		0.0%	-5.4%	17,249	156.7	86.4	37.1	17.4	98%	133%	113%	35%	57%	76%	NM	NM	NM
6515.TW	Winway	5,615.0	TWD	12,000.0	114%	O		0.9%	1.3%	6,339	108.2	NM	26.2	12.8	99%	132%	104%	44%	69%	84%	NM	NM	NM
6488.TWO	GlobalWafers	1,130.0	TWD	900.0	-20%	E		2.7%	1.7%	17,060	353.7	47.7	31.8	22.1	55%	50%	44%	12%	16%	21%	5.5	5.0	4.4
0522.HK	ASM Pacific	163.8	HKD	248.0	51%	O		1.8%	6.5%	8,328	63.1	37.3	24.0	17.4	73%	55%	38%	10%	15%	19%	3.6	3.3	3.0
688234.SS	SICC	102.5	CNY	127.1	24%	O		0.6%	NA	6,901	173.8	NM	NM	NM	NM	NM	NM	1%	3%	4%	6.9	6.7	6.4
ACMR.O	ACM Research	70.8	USD	115.0	62%	O		1.1%	-0.7%	4,645	84.5	25.0	18.3	12.9	104%	37%	42%	10%	13%	16%	2.2	2.0	1.8
3680.TWO	Gudeng Precision	571.0	TWD	648.0	13%	O		0.0%	-6.1%	1,732	24.9	29.4	20.8	15.0	106%	42%	38%	15%	19%	22%	4.2	3.6	3.0
6187.TWO	AllRing Tech	1,525.0	TWD	2,000.0	31%	O		2.8%	1.1%	4,634	105.8	49.3	31.7	26.8	100%	56%	18%	35%	41%	38%	NM	NM	NM
Mean:		1.2%						1.7%				47.6	29.6	20.2	81%	65%	52%	22%	30%	36%	6.0	6.1	5.0
Median:		0.9%						1.1%				47.7	29.1	17.4	92%	56%	40%	17%	21%	22%	5.5	5.8	5.2

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Data as of October 8, 2026.

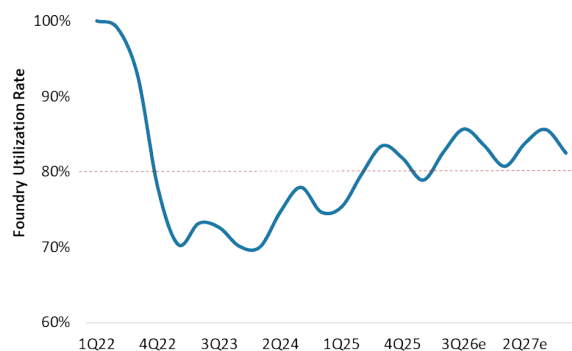
Valuation Comparison: Fabless, Power Semis, FPGA and Analog

Ticker	Company	Price 10/8/2026	Curr ency	Price Target	Upside/ Downside	Rating	Dividend Yield (%)	FCF Yield (%)	Market Cap (US\$M)	3M Avg. Daily Trading (US\$M)	P/E Ratio (x)			EPS Growth			ROAE			P/B Ratio (x)		
											2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e	2026e	2027e	2028e
Fabless																						
AI accelerators																						
688256.SS	Cambricon	981.0	RMB	1,700.0	73%	O	0.0%	-2.2%	85,623	2,021.5	106.8	43.9	31.7	89%	143%	39%	37%	61%	52%	37.0	21.1	13.3
9903.HK	Iluvatar	92.3	HKD	222.7	141%	O	NA	NA	0	154.3	NM	NM	NM	NA	NA	NA	-4%	13%	20%	0.0	0.0	0.0
688041.SS	Hygon	220.5	RMB	430.0	95%	O	0.1%	NA	70,641	971.8	108.3	72.5	46.5	86%	49%	56%	19%	24%	29%	19.2	15.7	12.2
688802.SS	MetaX	467.6	RMB	700.0	50%	E	0.0%	NA	25,990	243.7	NM	355.1	203.3	-103%	1844%	75%	0%	4%	6%	14.2	13.6	12.8
Smartphone/Wireless Semis																						
2454.TW	MediaTek	4,690.0	TWD	6,188.0	32%	O	19.2%	0.9%	231,795	1,161.9	64.0	31.9	13.7	11%	100%	133%	26%	40%	60%	15.3	10.6	6.5
603501.SS	OmniVision	79.0	CNY	111.0	41%	E	0.5%	2.1%	13,228	284.1	23.7	16.3	13.3	0%	45%	23%	14%	18%	19%	3.0	2.6	2.2
300782.SZ	Maxscend	64.7	CNY	58.00	-10%	U	0.1%	-2.2%	4,794	146.4	NM	173.4	49.6	113%	-132%	250%	-7%	2%	8%	3.7	3.7	3.4
603160.SS	Goodix	51.6	CNY	58.0	13%	U	0.7%	-1.4%	3,297	44.2	43.2	38.2	34.5	-35%	13%	11%	6%	6%	7%	2.5	2.3	2.2
6462.TWO	Egis	100.5	TWD	105.0	4%	E	1.0%	-16.3%	290	1.7	NM	18.7	NM	-88%	-462%	NA	-1%	5%	NA	1.0	1.0	NA
IP/Design Service																						
3661.TW	Alchip	4,345.0	TWD	5,888.0	36%	O	1.0%	5.1%	11,133	281.8	24.6	19.2	12.0	161%	28%	61%	32%	33%	44%	NM	NM	NM
301269.SZ	Empyrean Technology	85.5	CNY	95.0	11%	E	0.1%	NA	6,475	125.8	NM	NM	NM	251%	159%	68%	4%	10%	15%	NM	NM	NM
3443.TW	GUC	8,360.0	TWD	6,288.0	-25%	O	0.8%	4.7%	35,376	346.3	167.3	60.5	39.9	79%	176%	52%	44%	74%	68%	NM	NM	NM
6531.TW	AP Memory	980.0	TWD	1,666.0	70%	O	1.2%	4.1%	5,008	95.0	55.4	27.7	15.0	129%	100%	84%	22%	36%	49%	11.1	8.8	6.4
PC/Cloud/Display Semis																						
3034.TW	Novatek	531.0	TWD	414.0	-22%	U	6.1%	14.3%	10,203	66.7	15.3	15.0	17.6	29%	2%	-15%	30%	28%	23%	4.3	4.1	4.1
2379.TW	Realtek	749.0	TWD	717.0	-4%	E	4.8%	5.0%	12,067	78.5	20.0	20.0	17.2	30%	0%	16%	34%	31%	34%	6.4	6.1	5.5
688008.SS	Montage	197.0	CNY	377.0	91%	O	NA	NA	16,663	1,480.8	55.7	29.8	21.0	NA	87%	42%	23%	29%	31%	5.1	3.8	2.8
5269.TW	ASMedia	1,490.0	TWD	1,482.0	-1%	U	4.1%	0.9%	3,513	27.5	16.2	16.7	15.7	25%	-3%	6%	18%	16%	15%	2.7	2.5	2.2
5274.TWO	Aspeed	17,485.0	TWD	24,000.0	37%	O	1.3%	1.4%	22,959	146.7	84.1	55.8	39.4	100%	51%	41%	86%	84%	78%	NM	NM	NM
4966.TWO	Parade	558.0	TWD	818.0	47%	O	2.9%	1.6%	1,372	13.3	18.4	12.9	11.1	-11%	42%	17%	10%	14%	15%	1.9	1.7	1.6
2458.TW	Elan Micro	132.0	TWD	250.0	89%	O	4.8%	9.6%	1,193	11.5	12.6	10.7	10.6	24%	18%	1%	27%	28%	26%	3.2	2.9	2.6
HIMX.O	Himax	13.3	TWD	17.4	31%	E	2.7%	8.3%	2,261	17.5	33.3	18.3	9.1	65%	82%	102%	8%	14%	26%	2.7	2.5	2.2
Analog IC																						
6415.TW	Silergy	462.5	TWD	388.0	-16%	U	1.4%	4.8%	5,667	50.7	45.6	28.1	21.8	59%	62%	29%	10%	15%	17%	4.4	3.9	3.5
300661.SZ	SG Micro	96.0	CNY	100.0	4%	E	0.3%	1.1%	8,271	229.2	66.6	45.3	34.5	57%	47%	31%	17%	21%	23%	NM	NM	NM
MCU																						
4919.TW	Nuvoton	155.5	TWD	95.0	-39%	U	0.8%	14.5%	2,061	35.4	NM	37.4	20.4	-70%	-455%	84%	-4%	13%	20%	5.2	4.5	3.7
688018.SS	Espressif	117.7	CNY	125.0	6%	O	NA	NA	3,806	105.9	NM	NM	NM	NA	NA	NA	12%	14%	15%	NM	NM	NM
FPGA																						
002049.SZ	Unigroup Guoxin	56.1	CNY	52.0	-7%	U	0.4%	1.5%	6,626	233.2	24.0	18.5	16.5	38%	29%	13%	14%	15%	15%	3.1	2.7	2.3
1385.HK	Shanghai Fudan	24.2	HKD	42.0	74%	O	0.1%	0.1%	6,905	13.0	16.6	17.0	15.4	415%	-2%	11%	18%	15%	14%	2.7	2.3	2.0
IC distributor																						
3702.TW	WPG	117.5	TWD	160.0	36%	O	7.2%	-27.2%	6,230	37.8	9.2	9.4	8.9	115%	-2%	5%	24%	21%	20%	1.9	1.8	1.7
3036.TW	WT Micro	215.5	TWD	349.0	62%	O	4.4%	-21.4%	8,611	58.7	8.7	7.7	6.8	107%	13%	13%	25%	25%	24%	2.0	1.8	1.6

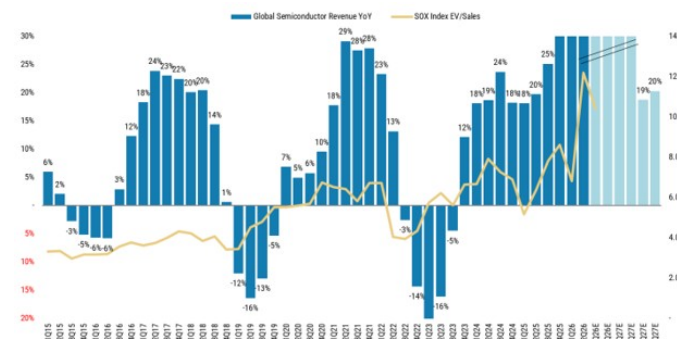
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates. Note: Data as of October 8, 2026.

Broader Semiconductor Cycle

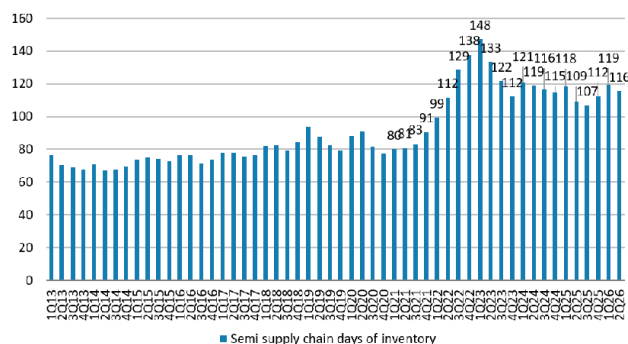
Logic semi foundry utilization could reach 80% in 2H26



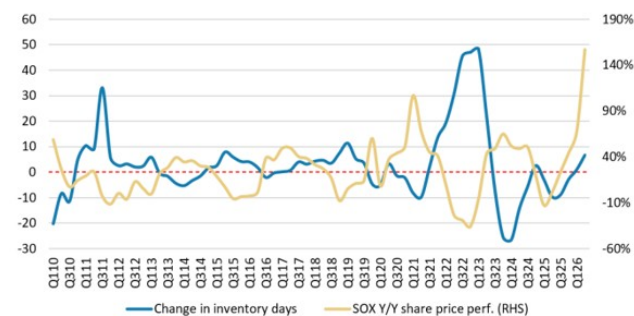
Excluding memory and NVIDIA's AI GPU revenue, non-AI semi growth is expected to decline in 2027



Semi supply chain – days of inventory as of 2Q26



Historically, when inventory days have gone down, the semi stock index has risen



Source: SIA, FactSet, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

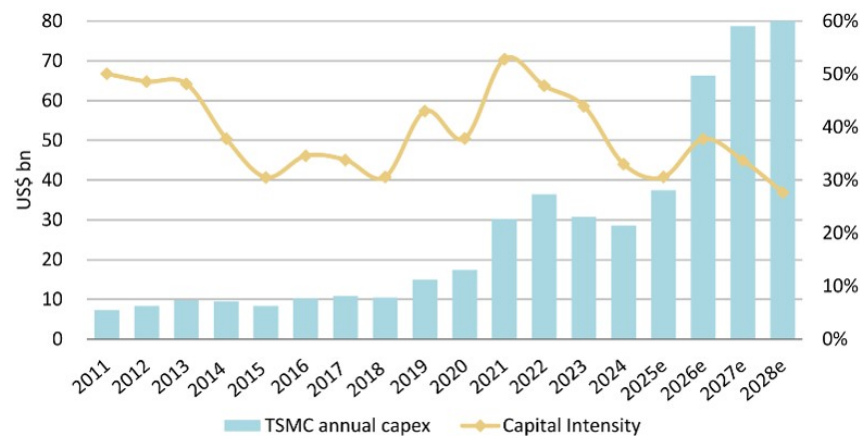
TSMC 3Q26 Earnings Preview

TSMC 3Q26 Earnings and 4Q26 Guidance Preview

3Q26 earnings and 4Q26 guidance preview

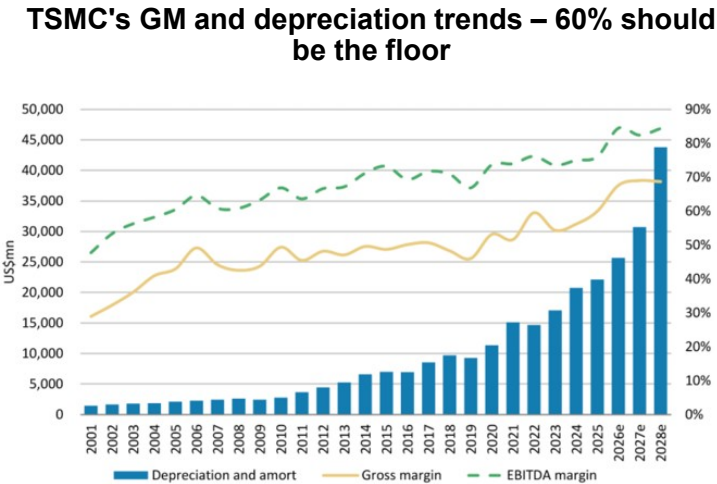
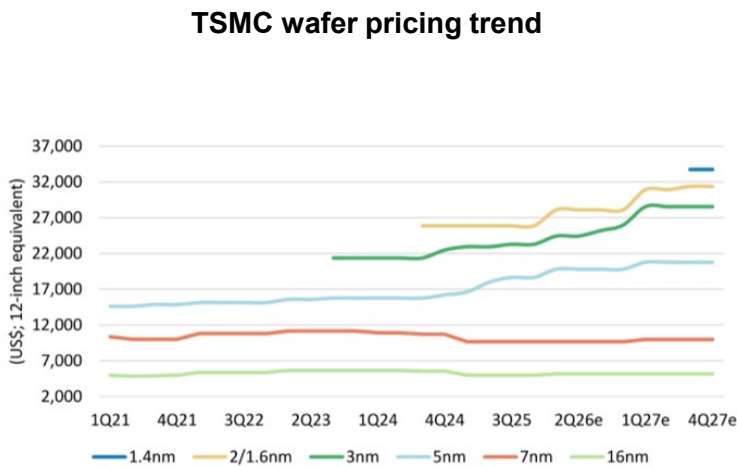
	3Q26				4Q26		
(NT\$ bn)	Guidance	Preliminary	MS Est.	Consensus	Guidance	MS Est.	Consensus
Revenue	US\$44.6 - 45.8bn		1,499	1,450		1,686	1,591
Q/Q	+12% at mid-point in USD			14.1%			9.7%
Y/Y				46.5%			52.1%
GM (%)	65.0% - 67.0%		67.5%	66.5%		68.6%	66.4%
OpM (%)	56.0% - 58.0%		60.8%	58.2%		62.2%	58.4%
EPS (NT\$)			30.74	28.28		35.25	31.23

TSMC's capex intensity should drop significantly as a result of robust revenue growth



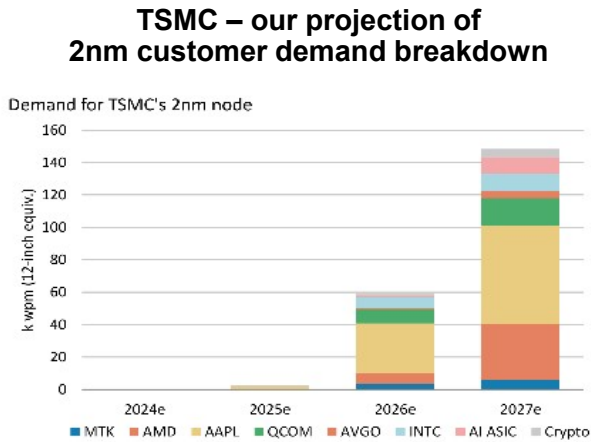
Source: Company Data, FactSet, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC Could Hike Leading-edge Wafer Price by 5-10% in 2027

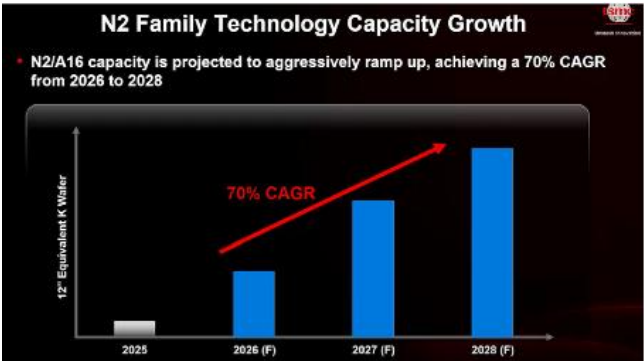


Source: Company Data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC 2nm Demand Still Remains Strong

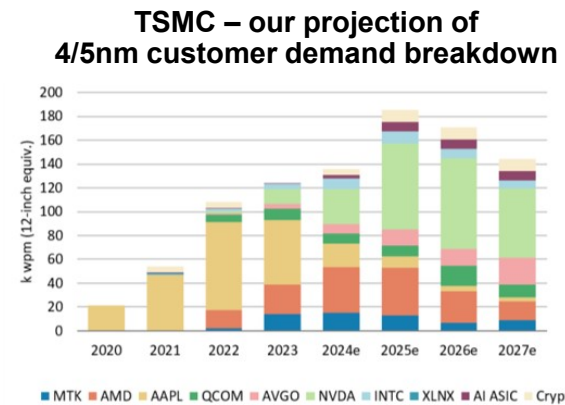
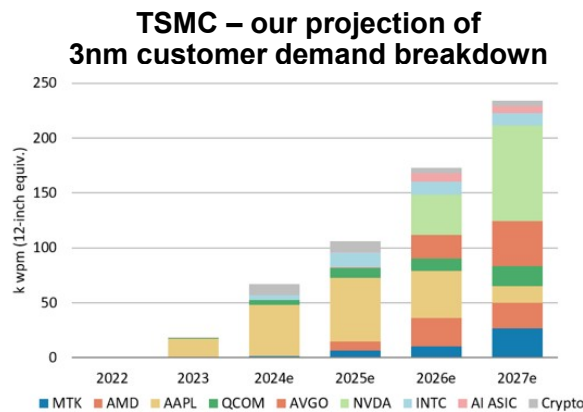


TSMC suggests N2 capacity to achieve 70% CAGR in 2026-2028e

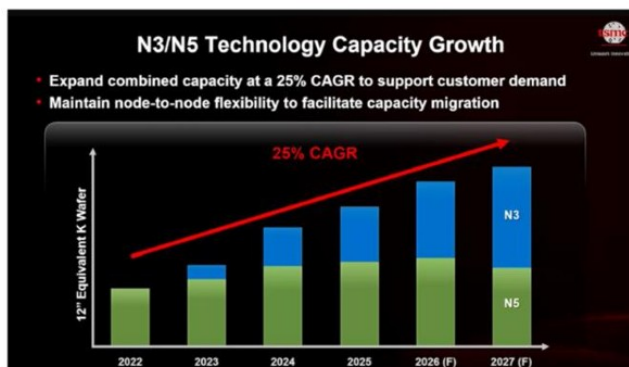


Source: Visible Alpha, Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

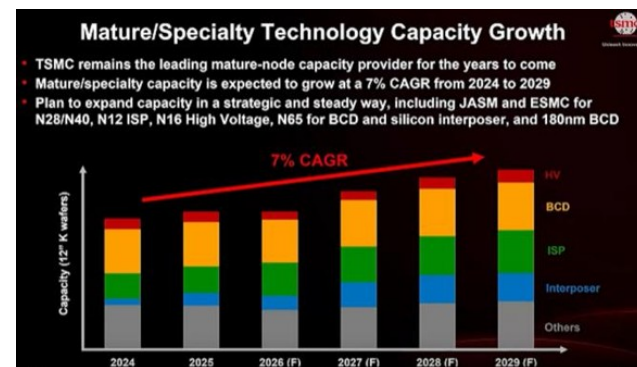
TSMC N5 Capacity to Drop in 2027 while N3 Should Keep Increasing, Similar to our Demand Forecast



TSMC indicates that N5 capacity will drop in 2027 while N3 capacity should keep increasing



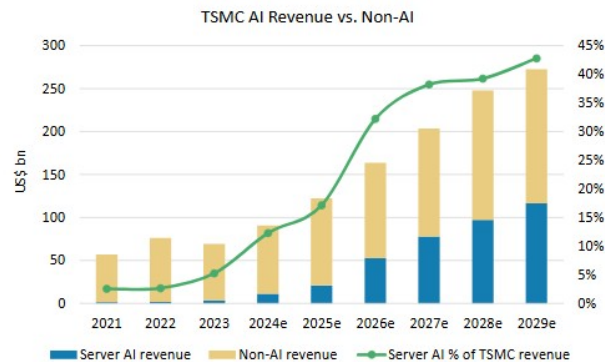
TSMC mature/specialty capacity is expected to grow 7% CAGR from 2024-2029



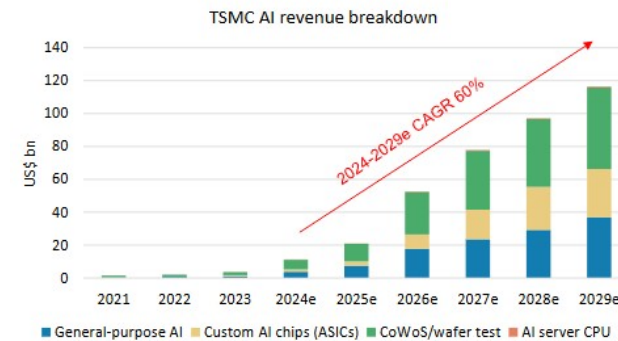
Source: Visible Alpha, company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

TSMC – AI Semis Revenue Could Reach 60% from 2024 to 2029e

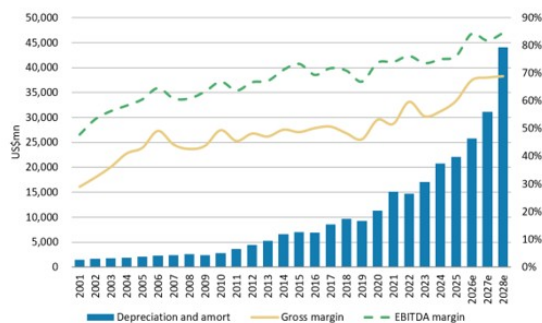
TSMC – AI semis revenue could account for >30% of 2026e revenue



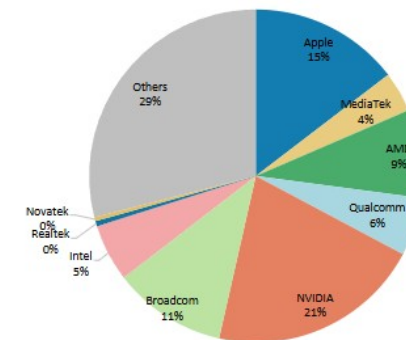
TSMC – AI semis revenue breakdown



TSMC – margin expansion



TSMC – 2026e revenue breakdown by customer

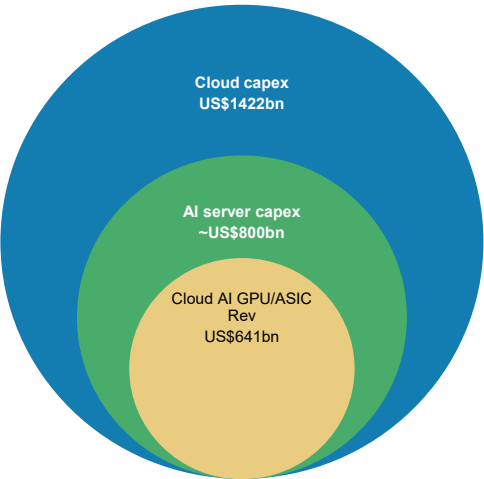


Source: Company data, Morgan Stanley Research (e) estimates

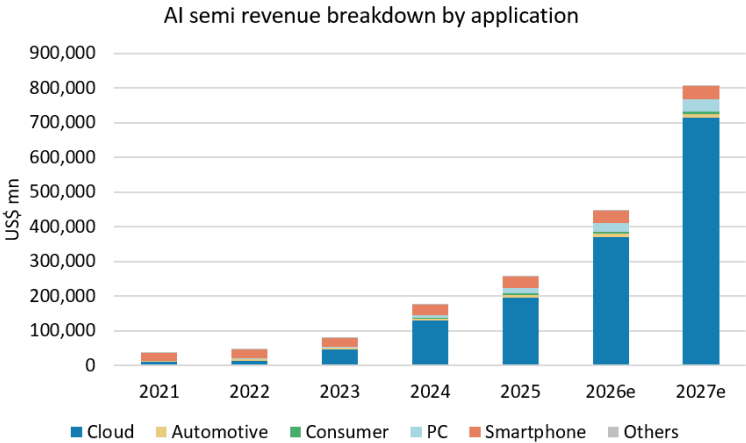
AI Semi Supply Chain and Niche Memory

Global Semi Industry Market Could Reach US\$2.5tn in 2027e, with AI Semis Contributing 1/3

Our supply chain data-driven bull case assumption is that cloud AI Semi TAM could grow to US\$641bn in 2027e



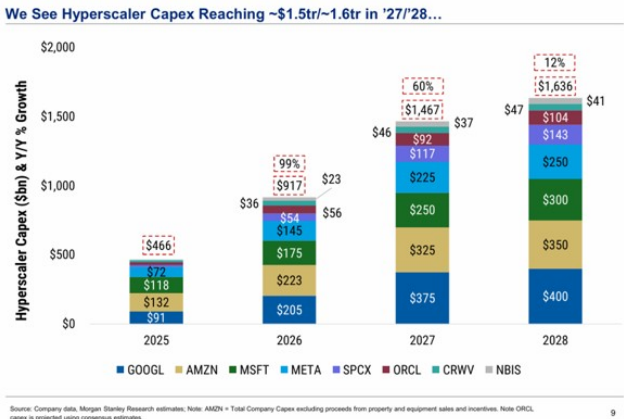
We expect AI semi TAM to reach ~US\$809bn in 2027e



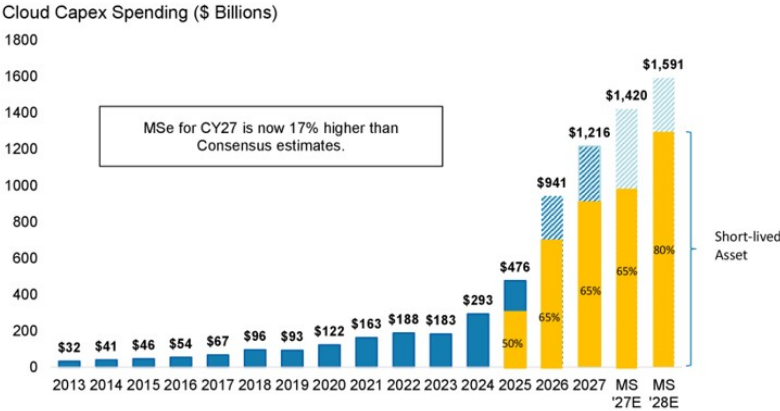
Source: Nvidia, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Deceleration of Cloud Capex From Major CSPs into 2028?

Hyperscaler Capex breakdown by major CSP: in aggregate, total 12% growth seen for 2028



Morgan Stanley cloud capex tracker: we estimate nearly US\$1.6tn of cloud capex in 2028 (top 14 listed global CSPs; no sovereign AI)



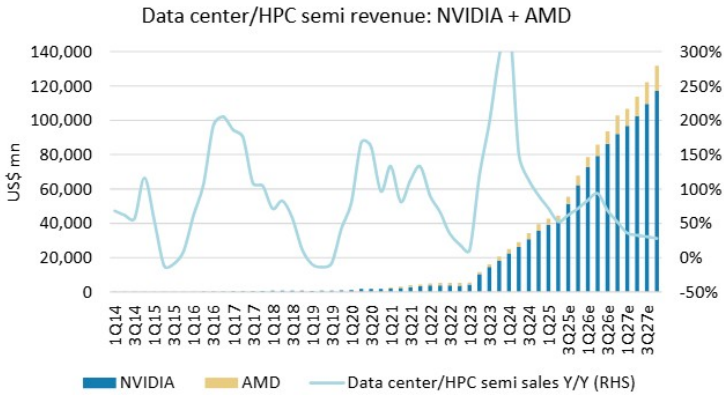
Source: FactSet, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are top-down and from our US tech team.

GPU Vendor P/E Multiple Might Catch up After Delivering Strong Revenue

P/E multiple trend of AI semis



We still expect AI chip revenue to rise QoQ

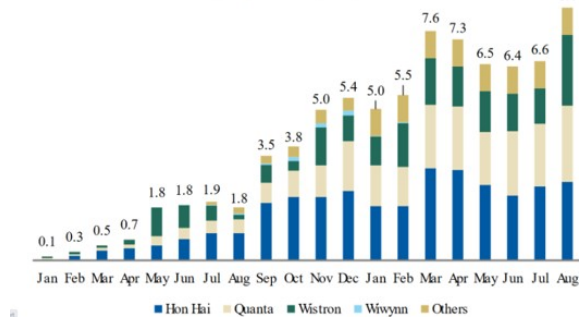


Source: FactSet, Company data, Morgan Stanley Research. Note: Alternative AI semis group: AMD, Alchip, Andes, Marvell, Broadcom. AI semi enablers group: TSMC, Synopsys, Cadence, ASML, BESl, Ibiden, KYEC, Advantest.

NVIDIA NVL72 Rack Forecasts

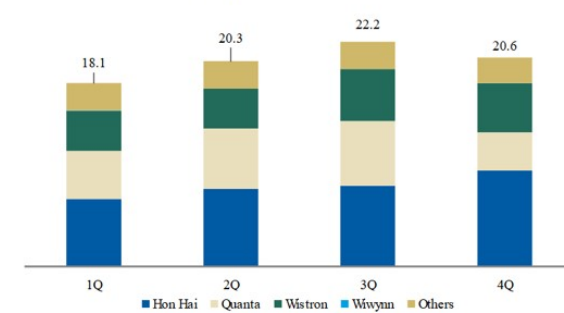
GB200/300 NVL72 Rack Ramp Monthly Trajectory

GB200/300 NVL72 racks by major ODMs (000s)

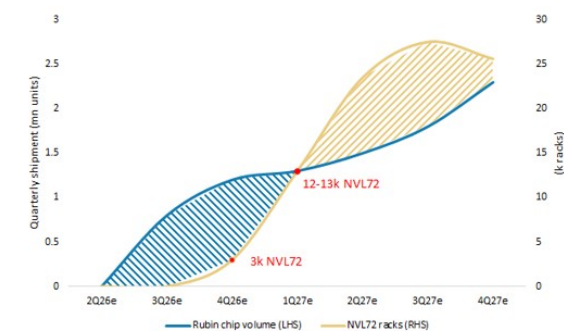
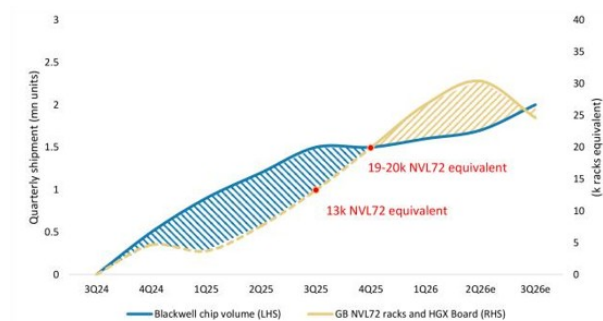


Industry-wide GB200/300 NVL72 Rack Split (2026)

GB200/300 NVL72 racks by major ODMs (000s)



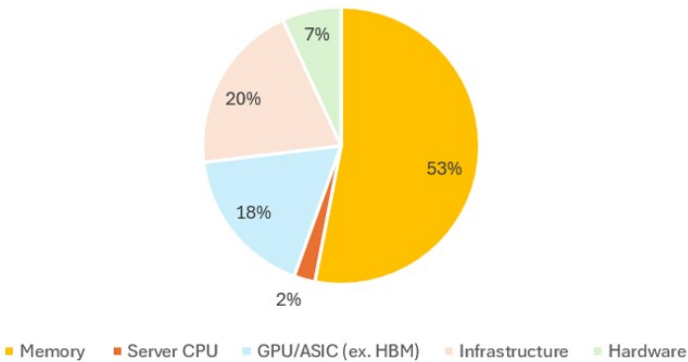
We estimate that TSMC produced 5.1mn chips in 2025, with full-year GB200 NVL72 shipments seen reaching 30k



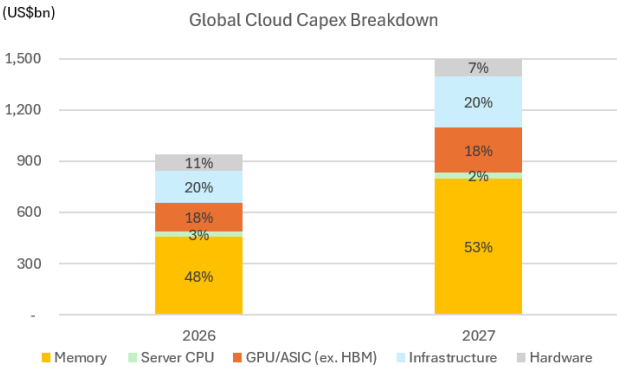
Source: Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: We think the actual rack deliveries to end-customers are lower than the numbers we have here because we include Wistron's computing tray (L10) rack equivalent number (without accounting for rack assembly and test times for L11). Estimates are from our Asia Pacific Tech team's downstream estimates.

Logic Cloud Accounts for Around 20% of Global Cloud Capex

2027 Cloud Capex Breakdown



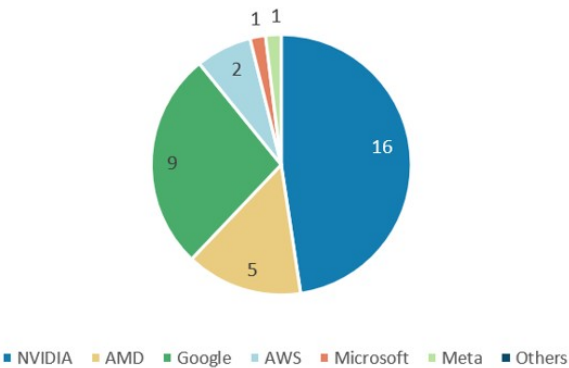
Global Cloud Capex Breakdown



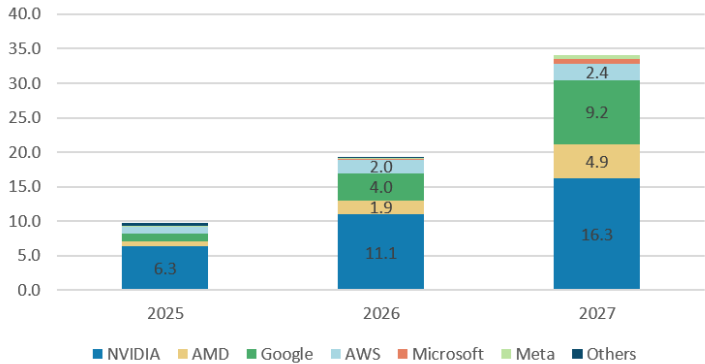
Source: Company data, Morgan Stanley Research estimates

Power Consumption Growth Remains Robust

2027 GPU/ASIC total power consumption (GW)



Global GPU/ASIC Total Power Consumption from TSMC CoWoS Shipment



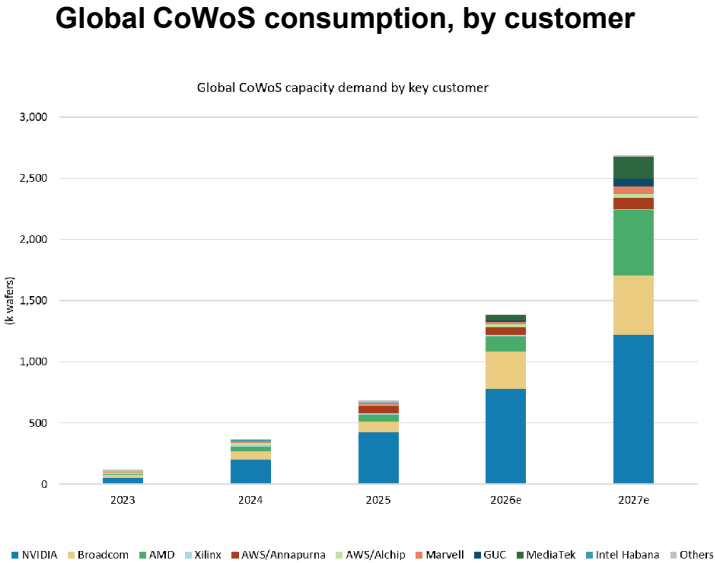
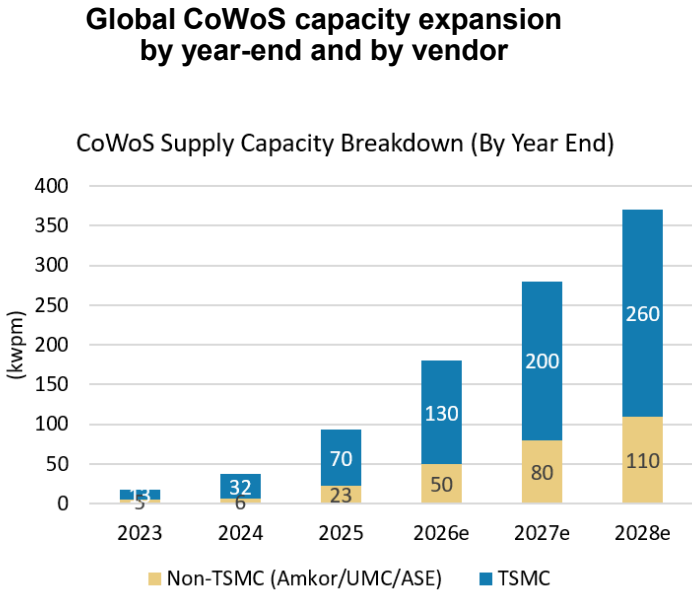
Source: Company data, Morgan Stanley Research estimates

Announced Power Deployment – Implications for TSMC

Announced Power Deployment Implication for TSMC					
GPU/ASIC Vendor	NVIDIA (Top US AI firm)	AMD (Top US AI firm)	Broadcom	AWS (Top US AI firm)	NVIDIA (SpaceX)
Power Deployed (GW)	10	6	3.5	2	8
Rack name	Vera Rubin NVL144	Helios	TPU	Trainium3 UltraServers	Vera Rubin NVL144
Power consumption per Rack (kW)	220	220	63	144	220
Rack Number (k units)	45	27	56	14	36
Chip Name	Rubin GPU	MI455 GPU	TPUv7 (Ironwood) TPUv8i (Sunfish)	Trainium3 (4)	Rubin GPU
Chip Volume (k units)	3,273	1,964	3,571	2,000	2,618
Life Cycle for the project	3	3	5	8	1
Implied life-cycle CoWoS Volume (k wafers)	409	166	260	118	409
Implied life-cycle 2/3nm wafer Volume (k wafers)	260	95	190	107	208
2027 annual CoWoS demand (k wafers)	136	55	52	15	409

Source: Company data, Morgan Stanley Research; Note: Estimates are compiled using our Asia supply-chain checks. Note: We assume 64 units of TPU chips per rack with TPUv7 and v8i per chip TDP being ~980W.

TSMC May Expand CoWoS Capacity to 260kwpm by 2028 Given Ongoing Strong AI Demand



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

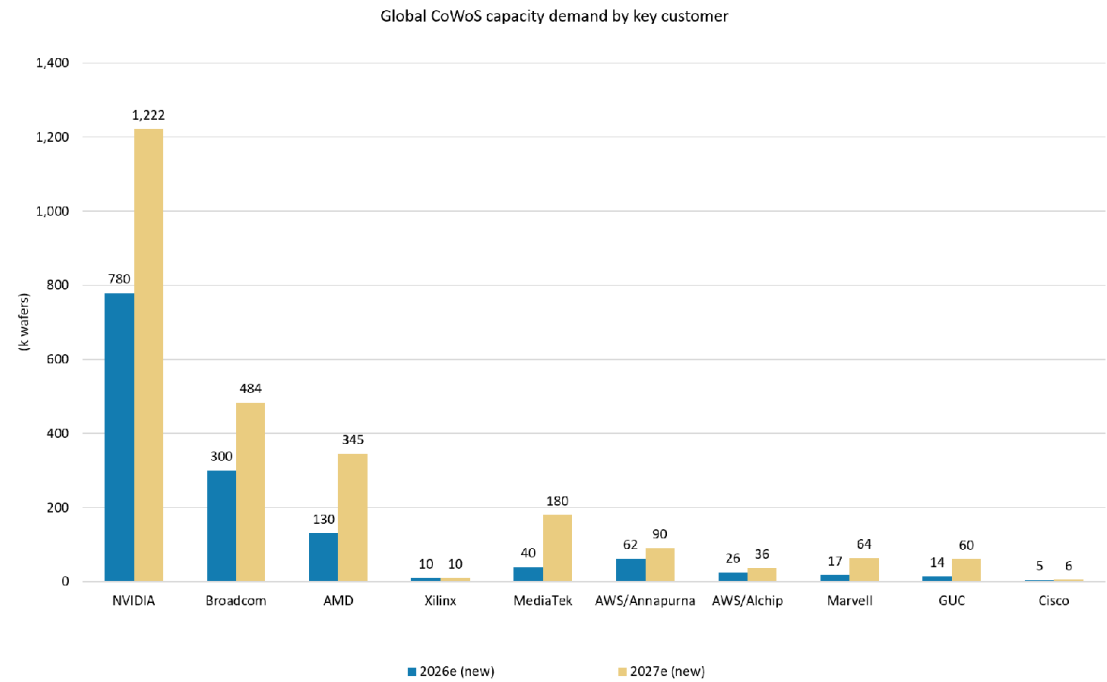
Global CoWoS Demand Breakdown

Factoring in Nvidia chip price hike starting in Q1 next year, 70%+ revenue growth in CY27 looks consistent with TSMC CoWoS implied chip shipments

(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
NVIDIA	53	200	425	780	1,222	45%	54%	62%	56%	49%
TSMC			390	680	1,090					
CoWoS-L			390	650	910					
CoWoS-S			0	20	50					
CoWoS-R			0	10	130					
Non-TSMC			35	100	132					
Amkor			35	100	132					
CoWoS-S			20	20	12					
CoWoS-R			15	80	120					
ASE/SPIL			0	0	0					
CoWoS-S			0	0	0					
CoWoS-R			0	0	0					
Broadcom	23	68	85	300	484	20%	18%	12%	22%	19%
TSMC			83	260	420					
CoWoS-L			0	15	60					
CoWoS-S			83	245	360					
ASE/SPIL			2	30	40					
CoWoS-L			0	0	0					
CoWoS-S			2	30	40					
Amkor				10	24					
CoWoS-S				10	24					
AMD	7	40	60	130	345	6%	11%	9%	9%	14%
TSMC			60	80	215					
CoWoS-L			0	70	210					
CoWoS-S			60	10	5					
ASE/SPIL			0	50	110					
CoWoS-L			0	50	110					
Amkor				0	20					
CoWoS-L				0	20					
(k wafer)	2023	2024	2025	2026e	2027e	2023	2024	2025e	2026e	2027e
Xilinx	3	10	10	10	10	3%	3%	1%	1%	0%
MediaTek				40	180				3%	7%
TSMC				40	180					
CoWoS-S				40	180					
AWS/Annapurna			60	62	90				4%	4%
TSMC			60	32	54					
CoWoS-R			60	32	54					
ASE/SPIL				30	36					
CoWoS-R				30	36					
AWS/Alchip	9	16	5	26	36	8%	4%	1%	2%	1%
Intel Habana	0	7	9	0	0	0%	2%	1%	0%	0%
Marvell	1	18	15	17	64	1%	5%	2%	1%	3%
TSMC				5	50					
CoWoS-L				5	50					
CoWoS-R			15	0	0					
ASE/SPIL				12	14					
CoWoS-R				12	14					
GUC	1	1	2	14	60	1%	0%	0%	1%	2%
Cisco		2	3	5	6		1%	0%	0%	0%
Others	20	10	15	10	12	17%	3%	2%	1%	0%
Total demand	117	372	689	1,394	2,509	100%	100%	100%	100%	100%

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

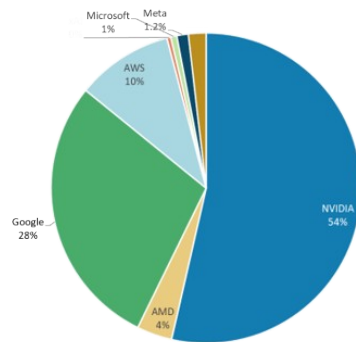
Global CoWoS Demand Set to Grow Robustly in 2027 at Each Major Customer



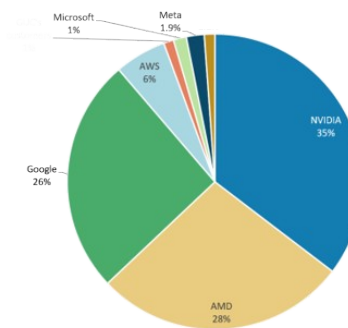
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are compiled using our supply-chain checks.

AI Computing Wafer Consumption Could Reach >US\$55bn in 2027, with NVIDIA Accounting for the Majority

AI semi wafer consumption, by customer, 2026e



AI semi wafer consumption, by customer, 2027e

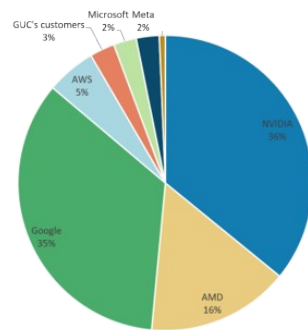


AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	Compute die size	Geometry	Compute die units	Wafer consumption (k wafers)	Wafer price (US\$)	Wafer revenue TAM (US\$ mn)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	850	4nm	2	44	23,042	1,024
	Vera CPU	250	23	5,750	850	3nm	1	228	27,300	6,229
	Rubin R200	740	8	5,920	850	3nm	2	470	27,300	12,827
	Rubin Ultra	130	8	1,040	850	3nm	2	83	27,300	2,253
AMD	MI350 series	5	12	60	110	3nm	8	2	27,300	48
	MI455	75	7	525	850	2nm	2	56	30,000	1,667
	MI450 (for Meta)	45	14	630	850	2nm	1	33	30,000	1,000
	MI500 (Arcadia)	10	4	40	850	2nm	2	4	30,000	127
	Venice CPU	210	21	4,410	155	2nm	8	176	30,000	5,292
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	700	3nm	2	14	27,300	371
	TPU v8i (Sunfish; AVGO)	330	12	3,960	800	3nm	2	296	27,300	8,075
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	800	3nm	2	269	27,300	7,341
	TPU v9 (Humufish; MediaTek)			400	850	2nm	4	63	30,000	1,905
AWS	Trainium 3	140	17	2,380	700	3nm	2	127	27,300	3,465
Microsoft	Maia 200	5	29	145	700	3nm	1	3.8	27,300	103
	Maia 300	50	11	550	850	2nm	1	29.1	30,000	873
Meta	MTIA 3 (Iris)	55	10	550	850	3nm	2	44	27,300	1,192
US AI Lab	Nexus (Titan 1)	25	13	325	750	3nm	4	45.5	26,000	1,183
	Titan 2	5	8	40	850	2nm	4	8.5	30,000	254
Others		18								
Total		2,454						1,996		55,230

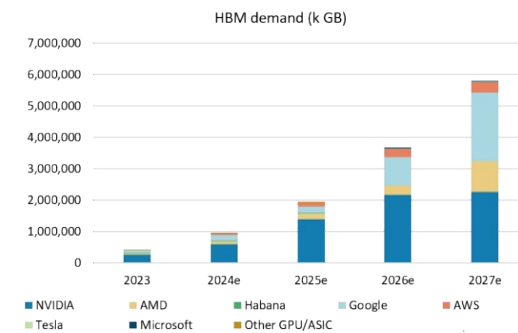
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Estimates are compiled using our supply-chain checks.

HBM Consumption in 2027e – Up to 44bn Gb

HBM consumption by customer in 2027e



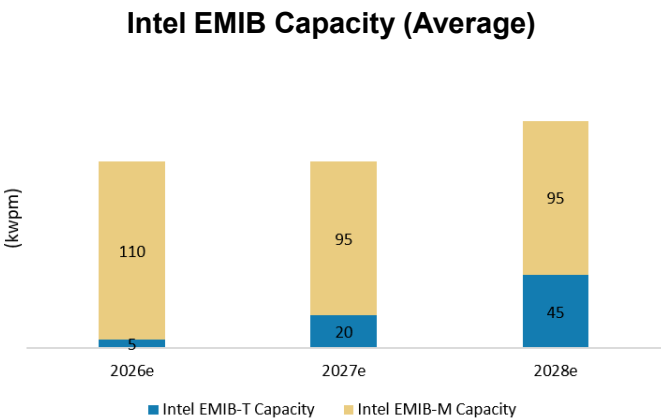
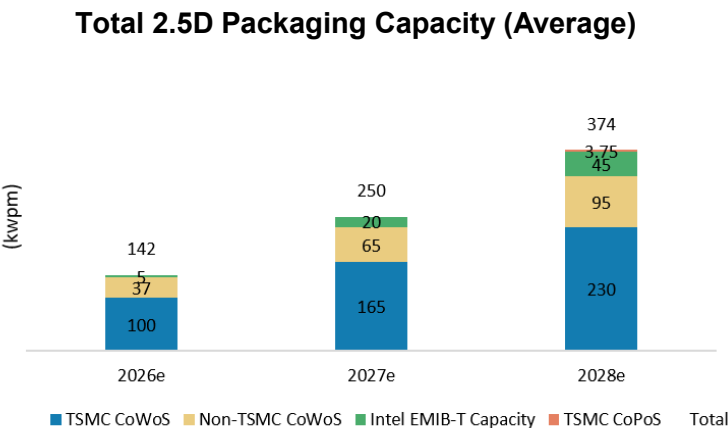
NVIDIA still consumes most HBM supply in 2027e



AI chip vendor	Product name	CoWoS capacity allocation (k wafers)	Chips per CoWoS wafer	Implied shipments (k)	HBM chip density (GB)	HBM chip units	Total HBM size (GB)	HBM generation	HBM vendor	Total HBM demand (k GB)
AI GPU (2027e)										
NVIDIA	B300	40	14	560	36	8	288	HBM3e 12hi	Hynix/Micron	161,280
	Vera CPU	250	23	5,750	36	8	288	HBM4 12hi	Hynix/Micron/Samsung	1,704,960
	Rubin R200	740	8	5,920	36	8	384	HBM4e 12hi	Hynix/Micron/Samsung	399,360
	Rubin Ultra	130	8	1,040	48	8	288	HBM3e 12hi	Samsung/Micron	17,280
AMD	Mi350 series	5	12	60	36	12	432	HBM4 12hi	Samsung/Micron	226,800
	Mi455	75	14	630	36	6	216	HBM4 12hi	Samsung/Micron	136,080
	Mi450 (for Meta)	45	4	40	48	16	768	HBM4e 12hi	Samsung/Micron	30,720
	Mi500 (Arcadia)	10	21	4,410						
AMD	Venice CPU	210								
AI ASIC (2027e)										
Google	TPU v7p (Ironwood; AVGO)	13	16	208	24	8	192	HBM3e 8hi	Hynix/Samsung	39,936
	TPU v8i (Sunfish; AVGO)	330	12	3,960	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	1,140,480
	TPU v8t (Zebrafish; MediaTek)	180	20	3,600	36	6	216	HBM3e 12hi	Samsung/Hynix/Micron	777,600
	TPU v9 (Humafish; MediaTek)	180	20	400	48	12	576	HBM4e 12hi	Samsung/Hynix/Micron	230,400
AWS	Trainium 3	140	17	2,380	36	4	144	HBM3e 12hi	Hynix/Samsung/Micron	342,720
Microsoft	Maia 200	5	29	145	16	4	64	HBM3	Samsung	9,280
	Maia 300	50	11	550	36	8	288	HBM4 12hi	Samsung	158,400
Meta	MTIA 3 (Iris)	55	10	550	36	8	288	HBM3e 12hi	Samsung/Hynix/Micron	158,400
US AI Lab	Nexus (Titan 1)	25	13	325	36	6	216	HBM4 12hi	Samsung/Hynix/Micron	70,200
	Titan 2	5	8	40	36	8	288	HBM4 12hi	Samsung/Hynix/Micron	11,520
Others		18								
Total		2,454								5,615,416
Total HBM demand (mn Gb)										44,923

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Total 2.5D Packaging Capacity Could Still Reach 50% YoY Growth in 2028



Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

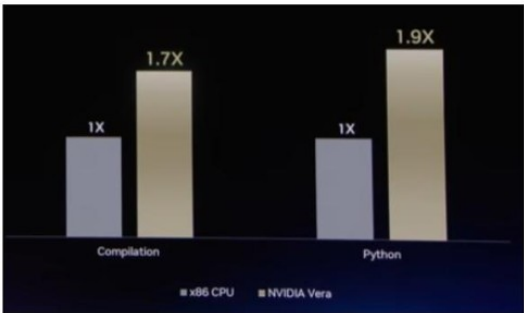
Total 2.5D/3D Packaging Solution Comparison

	CoWoS	CoPoS	SolC	EMIB
Packaging solutions	2.5D Wafer level	2.5D Panel level	3D Chip/die stacking	2.5D Substrate level
Performance (data bandwidth)	High	Low	Highest	Low
Efficiency for large chip (use of the substrate area)	Low	High	NA	High
Production sweet spot of the chip size	<9.7x reticle size	>9.7x reticle size	>9.7x reticle size	>9.7x reticle size
Cost at 9.7x retical size (including yield issue)	High	Theoretical low	Highest	Theoretical low
Technology maturity (opposite of execution risk)	High	Low	Medium	Low
Thermal challenge	Medium	Theoretical low	Highest	Theoretical low
Overview	CoWoS excels in high-bandwidth 2.5D integration but is costly and capacity-limited	CoPoS targets large-scale, cost-efficient integration but is still maturing	SolC delivers the highest performance through 3D stacking at the expense of complexity and thermal constraints	EMIB offers a more cost-effective and modular alternative with some performance trade-offs
Strategic consideration from customers				
Pros	Reliable and time to market	Cost efficient with larger chip size	Breakthrough the planar constraints	Cost efficient with larger chip size
Cons	Higher Cost	Not currently available at TSMC	High cost with thermal issue	Execution Risk

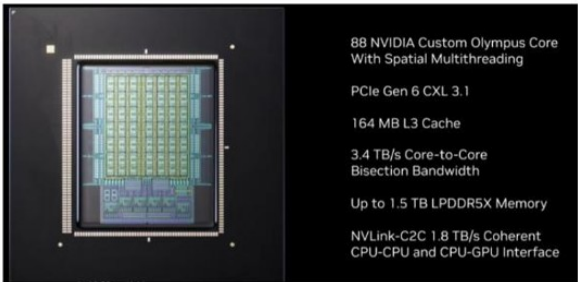
Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Nvidia’s Latest Vera CPU Could Support Robust Agentic CPU Demand

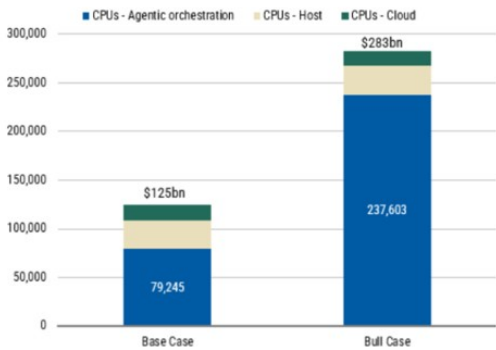
Vera performance could be 1.8x that of highest-performance x86 CPU



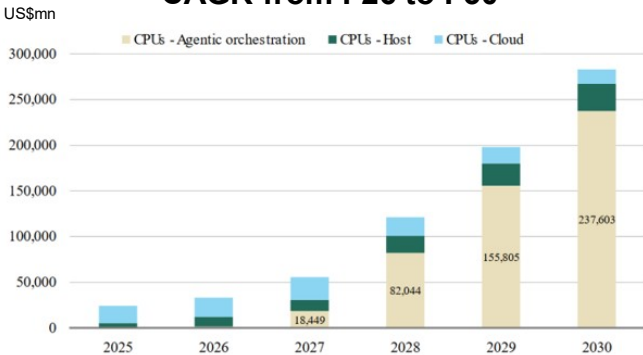
Vera CPU cores are not split across chiplets, enabling faster core-to-core connection



Our bull case implies a US\$238bn CPU orchestration TAM



In MS top-down model, we estimate the Agentic CPU TAM will expand at a 251% CAGR from F26 to F30



Source: Nvidia, Morgan Stanley Research (e) estimates from our Global team; see [Global Technology: Agentic AI – The Surge Begins \(11 May 2026\)](#)

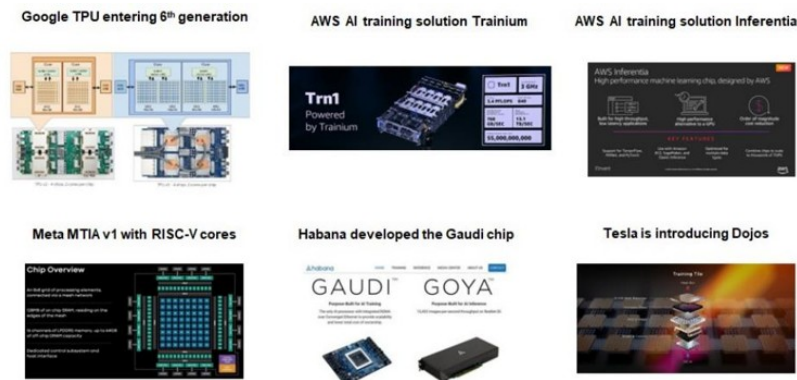
AI ASIC and Chip Testing

AI Semis: Now and the Future – “Prompts”

- Generative AI Demand Driver:
 - Killer apps
 - Competition
 - Sovereign
- Growth limitations:
 1. Budget
 2. Energy -> in the US
 3. Chip Capacity -> in China
 4. Regulation
- Semiconductor solutions:
 - Moore's Law
 - CoWoS/ SoIC
 - HBM
 - CPO
 - Custom chip
 - GaN HVDC 800V
- Growth perspective:
 1. Inference vs. Training AI semis
 2. Edge vs. Cloud AI semis
 3. Custom ASIC vs. AI GPU

CSPs Still Need Custom Chips, Even with NVIDIA Providing Powerful AI GPUs

Global CSP cloud AI custom chips



The volume upside of AI chips like Google TPU could depend on ABF substrate supply



Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

Alchip Might Achieve US\$8bn Revenue for Trainium4 in 2028

Alchip 2028 Trainium revenue could reach US\$8bn

		2026e	2027e	2028e
Volume (k)	Trainium3	400	600	
	Trainium4			800
Chip ASP Recognized by Alchip (US\$k)	Trainium3	4.6	4.6	
	Trainium4			10
Trainium Revenue (US\$bn)	Trainium3	1.8	2.8	
	Trainium4			8
	Total Trainium Revenue	1.8	2.8	8
Total Alchip Revenue (US\$bn)		3.1	4.4	9.8
Trainium Revenue %		60%	63%	82%

Trainium Forecasts from Our Asian Research team

k Units	2023	2024	2025e	2026e	2027e	2028e
Trainium1/Inferentia2	300	600	30			
Trainium2		300	1,200			
Trainium3			85	1,700	2,380	-
Trainium4					unknown	2,500
Total	300	900	1,315	1,700	2,380	2,500

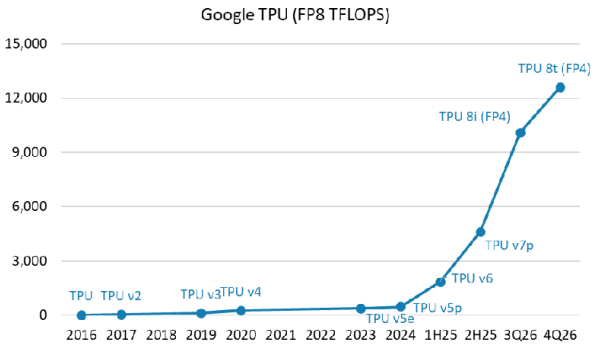
Source: AWS, Google, Meta, Intel, Tesla, Morgan Stanley Research (e) estimates.

More ASIC Projects Are Coming, According to Each CSP Plan

TPU Forecasts from MS Asia Research team

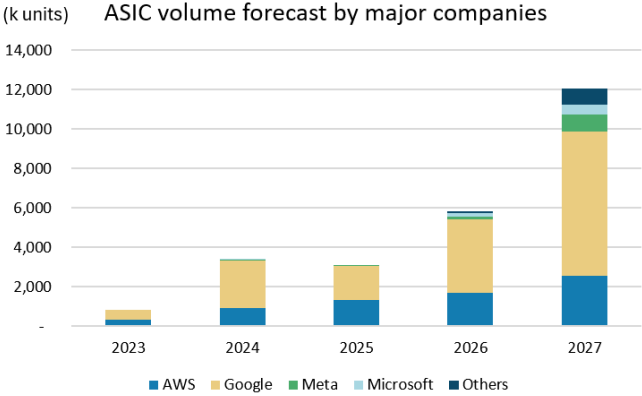
k Units	2023	2024	2025e	2026e	2027e	2028e
v5	500	2,400	250			
v6 (Trillium)			1,200			
v8t (Zebrafish; 3nm, by MediaTek)				500	3,000	2,000
v9 (Humufish; 2nm EMIB-T, by MediaTek)						3,000
v10 (Icefish; 1.4nm, by MediaTek)						unknown
Total	500	2,400	1,750	3,300	7,500	>9000

Google’s TPU performance is growing rapidly



Source: Company data, Morgan Stanley Research (e) estimates

ASIC volume forecast by major companies



MediaTek TPU Revenue Could Reach US\$70bn by 2029 Even Under a COT Model

We believe MediaTek would still at least handle the packaging and I/O die for TPU v10

MediaTek/Intel EMIB-T advanced packaging	
Google	KGD (Known Good Die) /Compute die
Memory Suppliers	HBM Chip
MediaTek	Electrical I/O Die
Marvell or others	Memory I/O
GUC or others	3D IC/CPO

We expect MediaTek 2029 TPU revenue to reach US\$70bn

		2027e	2028e	2029e
Volume (mn)	TPU v8t (Zebrafish)	3	2	
	TPU v9 (Humufish; EMIB-T)		3	4
	TPU v10 (Icefish)			1
	2nd Customer Subsystem		0.1	0.5
Chip ASP Recognized by MediaTek (US\$k)	TPU v8t	4.5	4.5	
	TPU v9; EMIB-T		13	13
	TPU v10			18
	2nd Customer Subsystem		25	25
Revenue (US\$bn)	TPU v8t	13.5	9	
	TPU v9; EMIB-T		39	52
	TPU v10			18
	Total TPU Revenue	13.5	48	70
	2nd Customer Subsystem		2.5	12.5
Total MediaTek Revenue (US\$bn)		35	75	NA
AI ASIC Revenue %		38%	68%	NA

Source: Company data, Morgan Stanley Research (e) estimates

KYEC Benefits from Strong AI GPU, TPU and CPU Growth

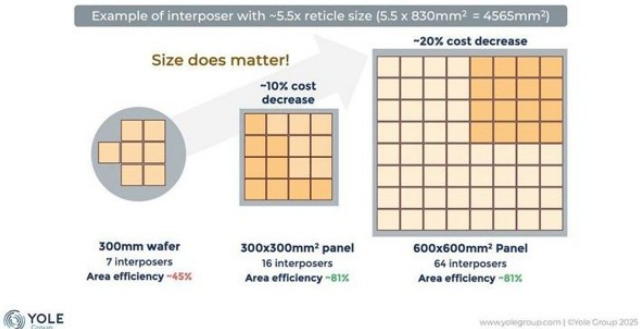
AI Revenue Forecast for KYEC

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26e	3Q26e	4Q26e	2025	2026e	2027e
Units (k)											
H-series	650	200	250	500	400	400	600	600	1,600	2,000	
B-series with GDDR7	-	-	300	500	250	250	250	250	800	1,000	
B-series with HBM	1,000	1,100	1,400	1,500	1,500	1,500	1,600	300	5,000	4,900	560
R-series							800	1,200		2,000	6,500
G-series CPU									1,100	2,450	250
V-series CPU										2,500	5,750
Google TPU									1,750	3,680	4,500
Google CPU										1,200	4,000
Test time											
Per chip test time (seconds)											
H-series test time	350	350	350	350	350	350	350	350	350	350	350
B-series with GDDR7 test time			350	350	350	350	350	350	350	350	350
B-series with HBM test time	700	750	900	1,000	1,000	1,000	850	850	1,000	850	850
R-series test time							850	850		850	1,275
G-series CPU test time									300	300	300
V-series CPU test time										700	700
Google TPU blended test time									186	340	450
Google CPU test time										300	400
Total test demand (k hrs)											
H-series	63	19	24	49	39	39	58	58	156	194	-
B-series with GDDR7			29	49	24	24	24	24	78	97	-
B-series with HBM	194	229	350	417	417	417	378	71	1,190	1,282	132
R-series					-	-	189	283		472	2,302
G-series CPU									92	204	21
V-series CPU										486	1,118
Google TPU									90	348	563
Google CPU										100	444
Test hour rate (for the largest customer including burn-in)	170	170	170	170	170	170	180	190	170	178	200
Test hour rate (for TPU with self-bought testers)									200	250	280
Largest customer's AI revenue to KYEC (US\$ mn)	43.8	42.3	68.6	87.4	81.6	81.6	116.9	83.0	257.6	485.5	714.6
% of KYEC MSe Revenue (quarterly figures without CPU)	19%	15%	23%	27%	25%	23%	30%	18%	23%	32%	33%
TPU revenue to KYEC (US\$ mn)									18.1	86.9	157.5
% of KYEC MSe Revenue									1.6%	5.7%	7.3%
Google CPU revenue to KYEC (US\$ mn)										25.0	124.4
% of KYEC MSe Revenue										1.6%	5.7%

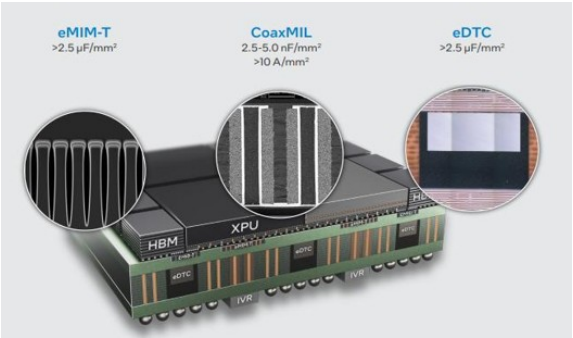
Source: Company data, Morgan Stanley Research (e) estimates

What About the Competition Between TSMC’s CoWoS and Intel’s EMIB?

TSMC CoWoS can support up to 9.5x
reticles, or four chips per wafer

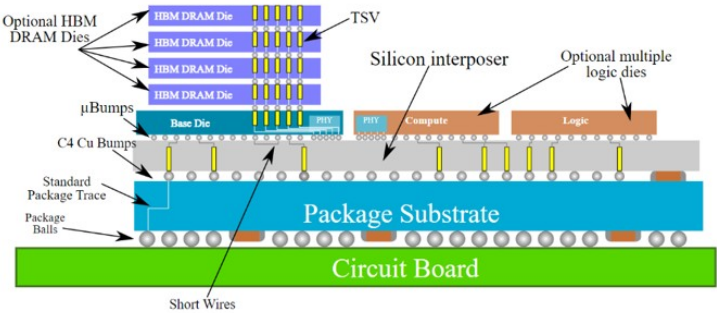


Intel EMIB eDTC: APMemory’s technology

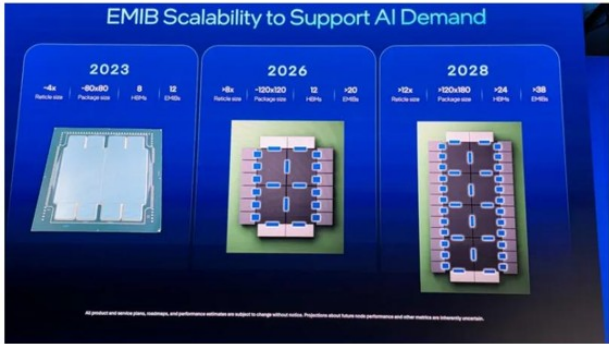


Source: Intel, Yole, Morgan Stanley Research

TSMC’s CoWoS-S (silicon interposer)
packaging architecture



Intel’s EMIB can support larger chips with more reticles
(>12) if its supply chain can execute well



CXMT in the Context of Global DRAM

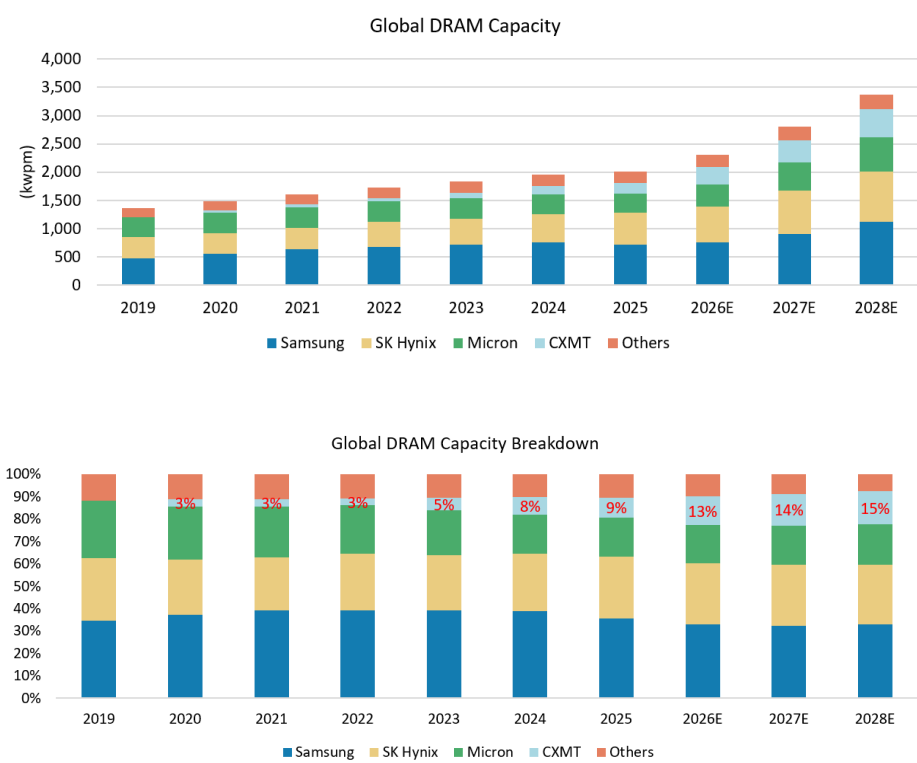
CXMT’s Process Node Roadmap and Specs Compared to Global Vendors

DRAM Process Node Roadmap															
CY	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Samsung	20nm		1X (18nm)		1Y (17nm)	1Z (16nm)	EUV	1a (14nm)	1b (12nm)			1c (10nm)		1d	
SK hynix		21nm			1X	1Y	1Z	1a		1b		1c		1d	
Micron		20nm		1X	1Y	1Z		1a		1β		1γ		1δ	
CXMT						G1 (19nm)			G3 (18nm)	G4 (16nm)	G4B (16nm)				G5 (15nm)

DDR5 Spec Comparison				
Manufacturer 16 Gb DDR5	Samsung 16 Gb DDR5	SK hynix 16 Gb DDR5	Micron 16 Gb DDR5	CXMT 16 Gb DDR5
Year First Revealed (Clock Speed, ex.)	2021DDR5-5600, DDR5-6000	2021DDR5-4800, DDR5-6000	2021DDR5-4800	2024DDR5-6000
Parent Product Examples	G.SKILL Trident Z5 DDR5 F5-5600U3636C16GX2-TZ5K (K4RAH086VB-BCQK)	SK Hynix 32 GB HMC88MEBUA81N DDR5 UDIMM PC5-4800B module (H5CG48MEBD-X014)	TeamGroup ELITE DDR516 GB UDIMM (MT60B2G8HB-48B:A)	Gloway 16GB×2 (32GB) DDR5-6000 Module(H2G08SC522-6AF)
DRAM Process / Geometry	D1y (17nm)	D1y (17nm)	D1z (16nm)	G4 (16nm)
Die Size (Scaled)	73.58 mm² (8.94 mm × 8.23 mm)	75.21 mm² (9.24 mm × 8.14 mm)	66.26 mm² (8.43 mm × 7.86 mm)	66.99 mm² (8.19 mm × 8.18 mm)
Bit Density	0.217 Gb/mm²	0.213 Gb/mm²	0.241 Gb/mm²	0.239 Gb/mm²
Peripheral Gate Process	HKMG (Gate-first)	Poly/W Gate	Poly/W Gate	Poly/W Gate

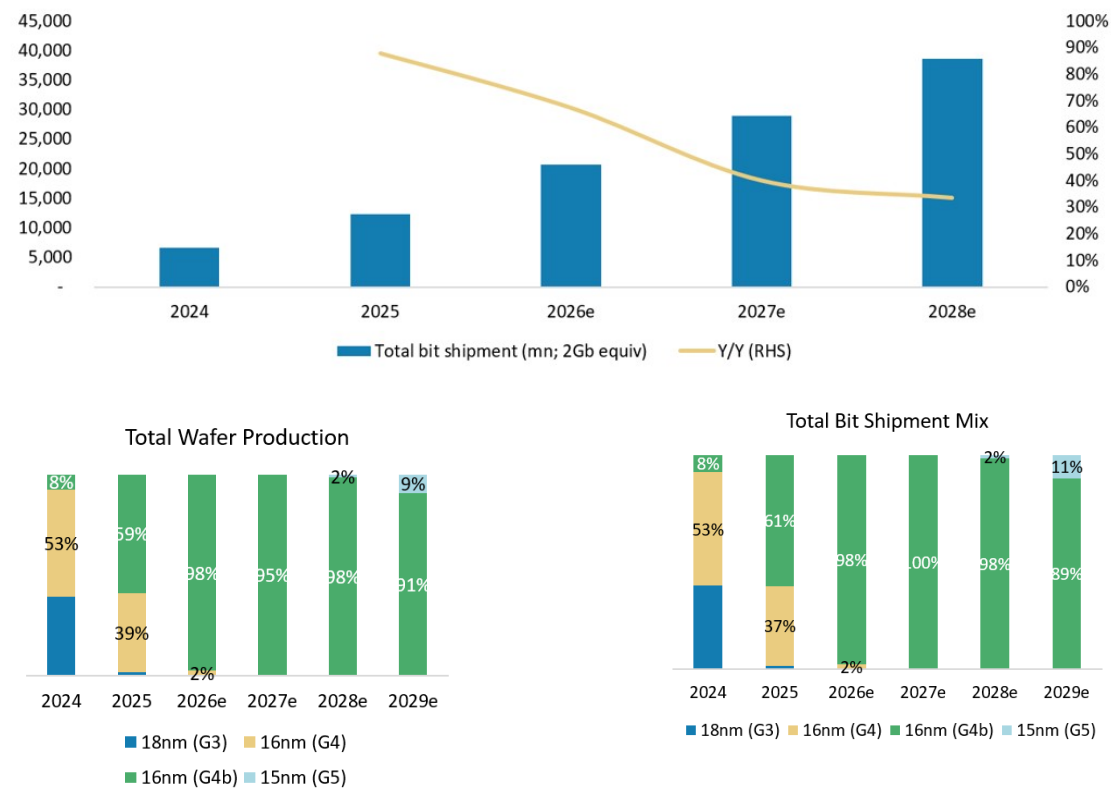
Source: Morgan Stanley Research

Global DRAM Capacity to Reach 3,374kwpm in 2028e, while CXMT May Account for 15% of That (Around 500kwpm by 2028)



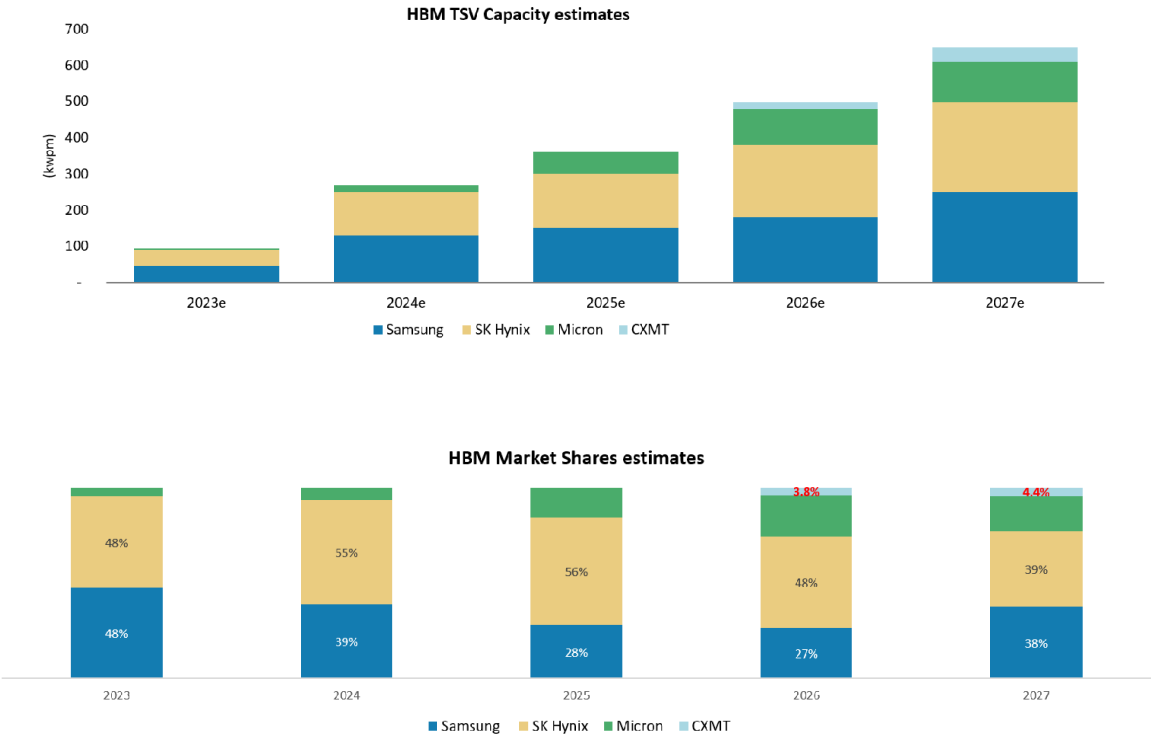
Source: Morgan Stanley Research; e = Morgan Stanley Research estimates

CXMT’s Total Bit Shipments to Rise at a 40% CAGR in 2025-28e, as Production Mix Shifts from G4b to G5 into 2028/29



Source: Morgan Stanley Research; e = Morgan Stanley Research estimates

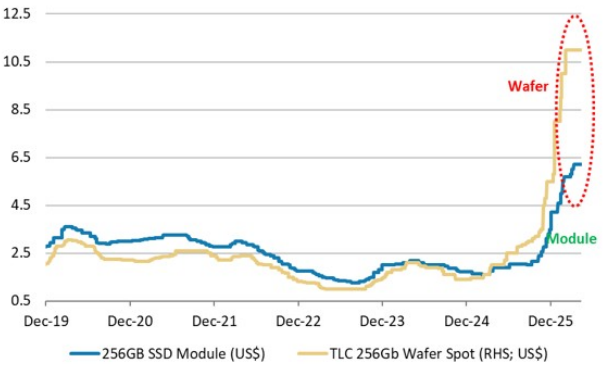
CXMT's HBM TSV Capacity to Reach 20kwpm in 2026e, 40kwpm in 2027e, and 70kwpm in 2028e; to Account for 3.8%/4.4% of the Total HBM Market in 2026e and 2027e in Terms of Bit Shipment



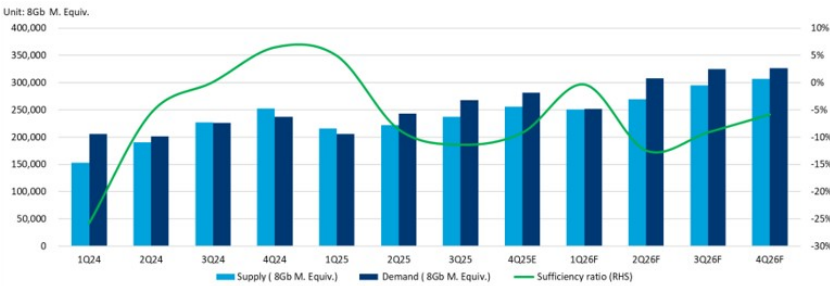
Source: Morgan Stanley Research; e = Morgan Stanley Research estimates

AI Storage Leads to NAND Shortage; We Also Expect NOR Flash Undersupply into 2026

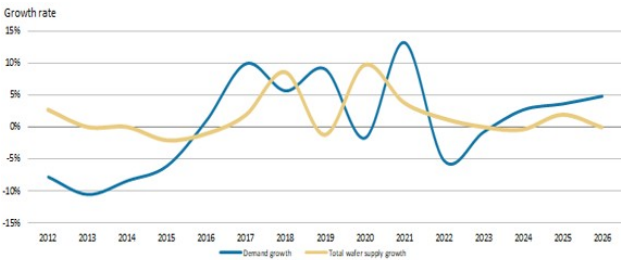
NAND wafer spot pricing and module prices



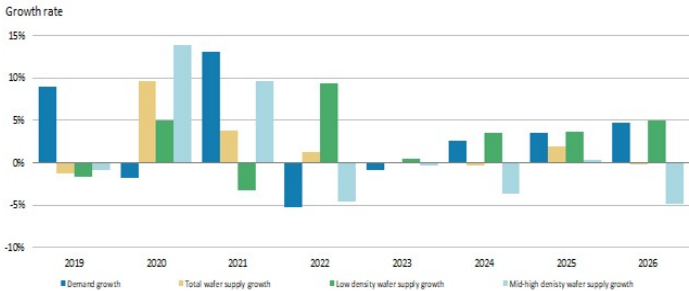
NAND supply and demand model



NOR flash demand and supply growth rates

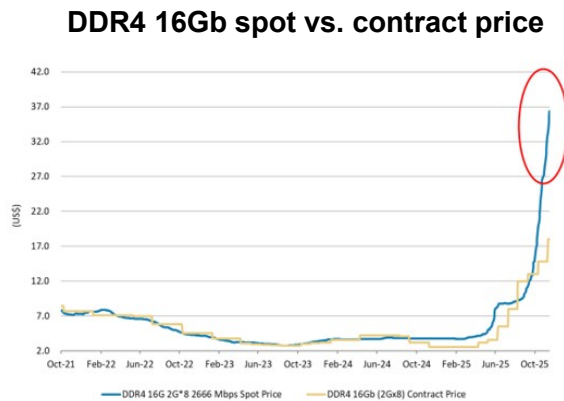


NOR flash demand and supply growth rates

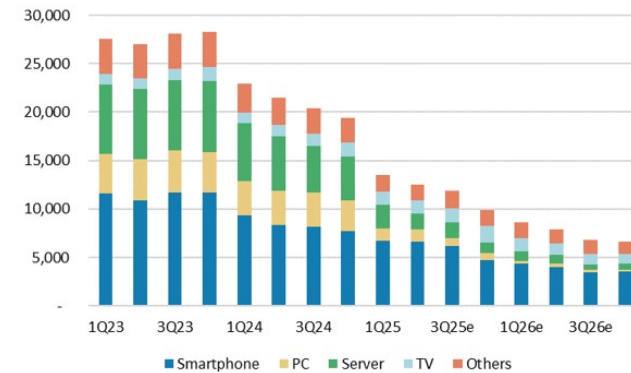


Source: ChinaFlashMarket, Morgan Stanley Research estimates.

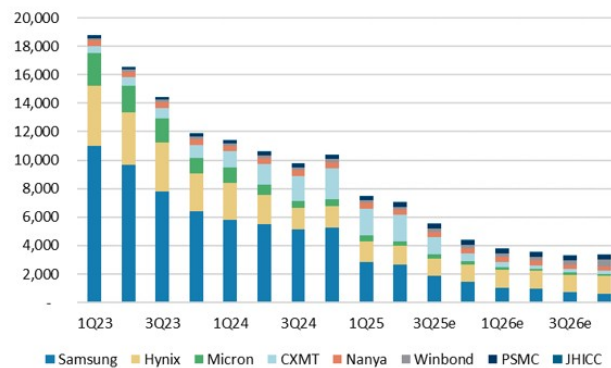
DDR4 Shortage to Persist into 2H26; Spot Price Capped



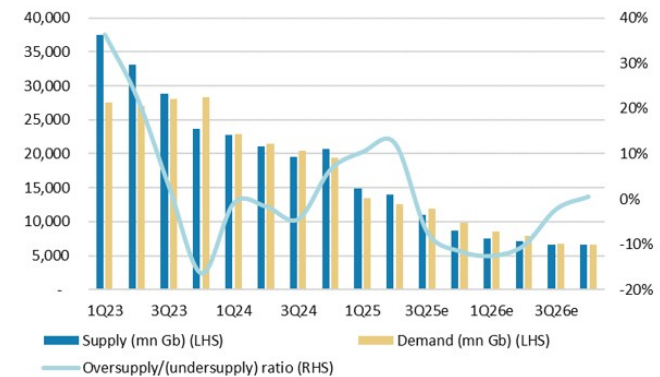
Quarterly demand breakdown, by product (mn Gb)



Quarterly supply breakdown (mn Gb)



Quarterly supply and demand summary

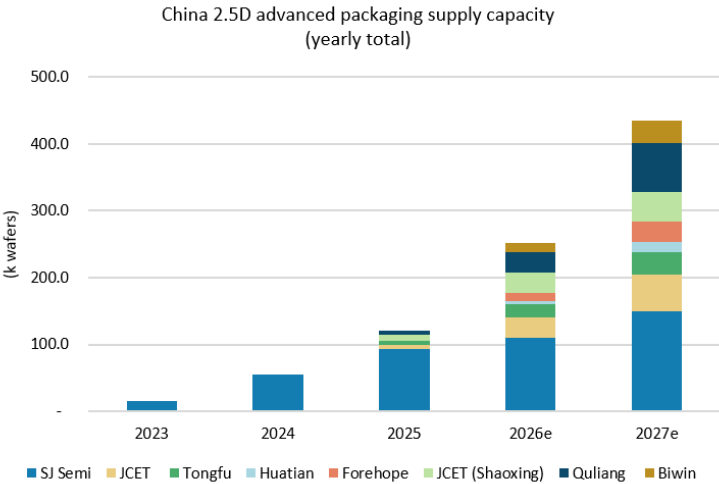


Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

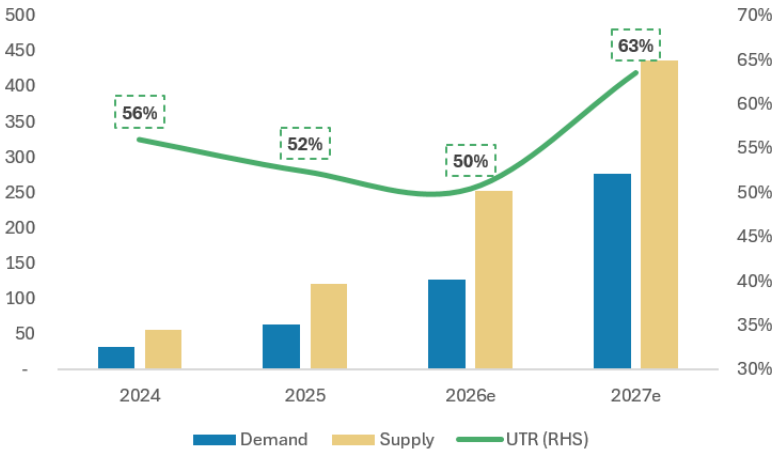
China Advanced Packaging

Favor Capex Beneficiaries and Share Gainers

We expect China's 2.5D packaging annual capacity to grow to 436k wafers in 2027



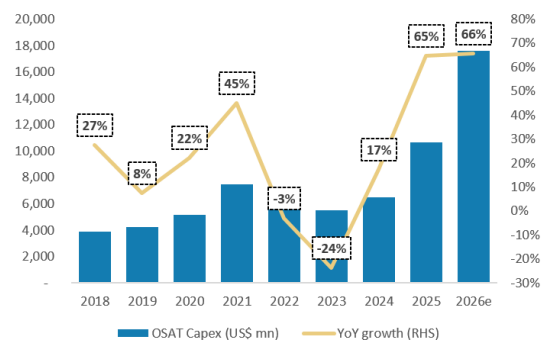
We expect moderate overcapacity of China's 2.5D packaging



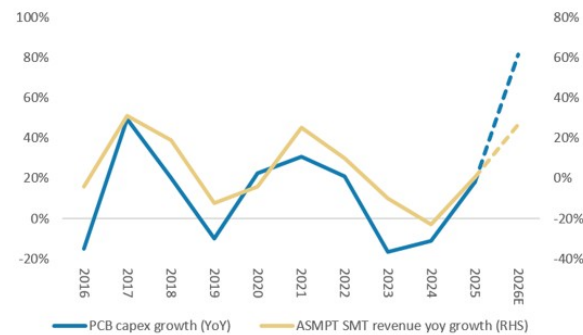
Source: Company data, Morgan Stanley Research estimates.

ASMPT (OW): Benefiting from Robust OSAT and PCB Capex; Other Drivers Emerging

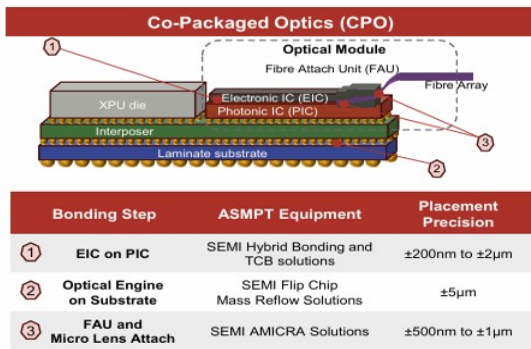
OSAT capex expected to grow ~66% in 2026 driven by both mainstream semi and advanced packaging



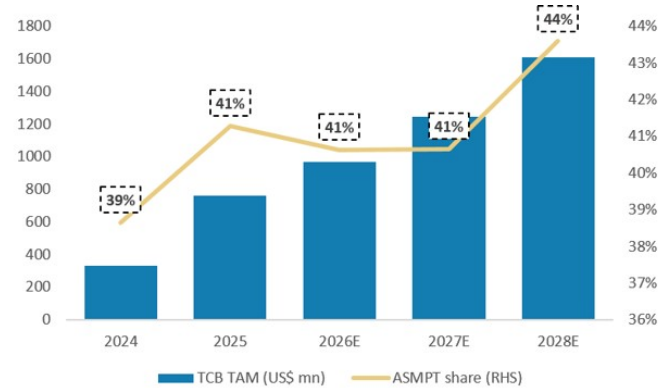
ASMPT SMT growth highly correlated with PCB capex



ASMPT Equipment Matrix – CPO Assembly Flow



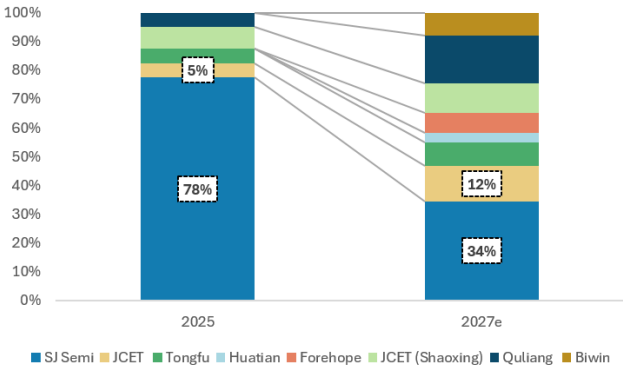
We expect ASMPT's share in TCB to be ~44% in 2028



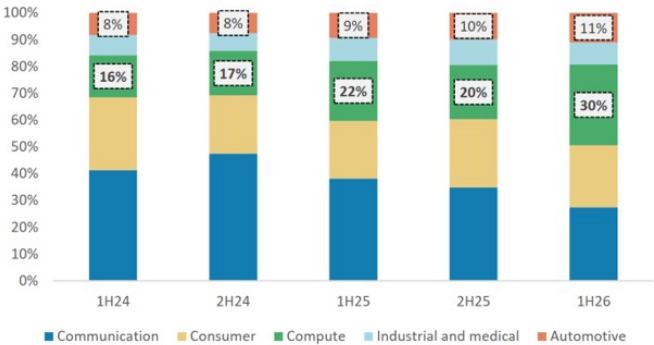
Source: Bloomberg, Company data, Morgan Stanley Research estimate

JCET (OW) : Gaining Share in AI GPU Packaging

JCET is likely to gain market share in 2.5D packaging



Computing and automotive mix continue to grow



Source: Company data, Morgan Stanley Research estimates.

CPO

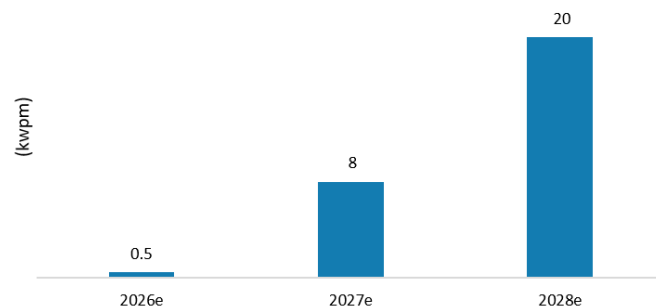
Larger OE TAM but Potentially Less Market Share for FOCI

Available OE estimates

	2026e	2027e	2028e
TSMC average PIC capacity (kwpm)	0.5	8	20
Die per wafer	648	648	648
Annual PIC production (mn)	4	62	156
SolC yield rate	50%	95%	95%
Optical engine production (mn)	2	59	148
Downstream assembly yield rate	20%	30%	50%
Actual Optical engine shipment (mn)	0.39	17.73	73.87

TSMC average PIC Capacity

TSMC average PIC Capacity Planning



Nvidia Revenue contribution to FOCI

	2026e	2027e	2028e
Rubin Ultra rack unit (k units)		5	20
Optical Engine content per Rubin Ultra rack		3072	3072
Scale-Up (NVLink switch/Quantum)		2304	2304
Scale-out (Spectrum)		768	768
Other scale-out switch optical engine consumption			
Spectrum-X (k units)	800	1,000	3,000
Quantum-X (k units)		1,000	3,000

Market Share Assumptions

FOCI	15%	10%	8%
Others (TFC, Corning, etc.)	80%	85%	85%

FAU ASP Assumptions

FAU ASP (6.4T; US\$)	220	220	220
FAU ASP (3.2T; US\$)	100	100	100
FAU ASP (1.6T; US\$)	45	45	45

FOCI Revenue contribution from NVIDIA

Revenue contribution (US\$mn)	9	186	597
Revenue contribution (NT\$mn)	261	5,568	17,914
FOCI CY Revenue (NT\$mn)	2,100	6,791	19,914
% of total revenue	12%	82%	90%

Source: Company data, Morgan Stanley Research estimates.

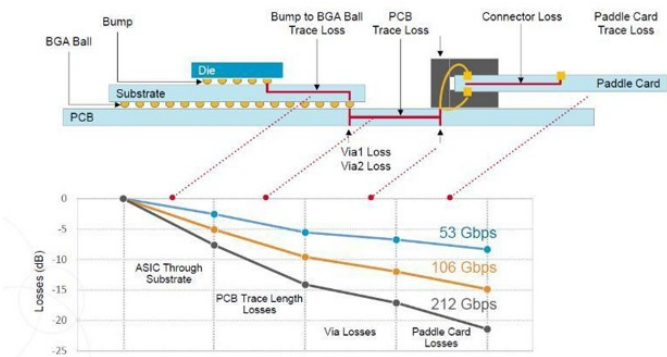
CPO Insertion Test Flow

Test Insertion	Insertion 1 (wafer-level)	Insertion 2 (wafer-level)	Insertion 3 (die-level)/3.5 (package-level)	Insertion 4 (module-level)		Insertion 5 (potential)
Test content	Electronic IC (EIC) & Photonic IC (PIC)	EIC die to PIC wafer (after SoIC)	Optical engine (die-to-die)	Insertion 4O CPO package (ASIC+OE) Optical Testing	Insertion 4E CPO package (ASIC+OE) Electrical Testing (2025-2026) Optical+Electrical Testing (starting 2027e)	CPO package (ASIC+OE) SLT?
Testing surface		 	 	 		
Key Product timeline	2025-2026 Scale-out CPO switch (Spectrum+Quantum) products 2027e onward: Scale-out CPO switch (Spectrum+Quantum) + Scale-up CPO switch products					
Testing details	EI. & opt. DC, (power, loss, dark current etc.)	E/O, O/E, O/O, high speed, S-parameters	Full calibration/DC, high-speed functional, optical loopback, alternatively: S-parameters	Optical Light Transmission Testing, optical loopback	BER testing, signal testing	Full system functional validation
Testing service provider	Foundry (Wafer-level) TSMC	Foundry (Wafer-level) TSMC	OSAT (Die/chip level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)	OSAT (Die/package level) SPIL (ASE Group)
Equipment and Consumable Vendors	PIC ATE tester: 1) Advantest + TEL + Formfactor 2) Chroma (underqualification) EIC tester: 1) Teradyne + TEL EIC Probe card: FormFactor	Wafer level testing: 1)Ficointec +Formfactor +Teradyne 2) Advantest+ MPI	Die-to-die prober: 1. TEL 2. MPI Optical engine E/O tester+laser reliability test+ELS source test head: Chroma Optical and Electrical Test Socket: Winway	CPO switch tester: Chroma Hypersocket: Winway	ATE tester: 1) Advantest (under qualification) 2)Teradyne FT Handler+ELS source test head+optical alignmnet stage : Hon Precision Hypersocket: Winway	CPO SLT tester: Chroma (under qualification) Coaxial socket/Hypersocket: Winway

Source: Company data, Morgan Stanley Research estimates.

CPO Could Help Enhance Data Transmission Speeds and Reduce Power Consumption

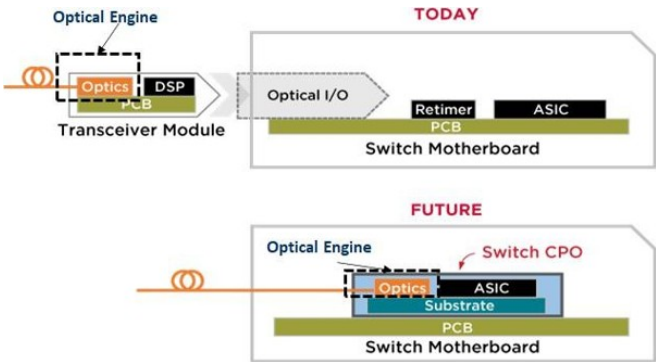
As transmission speeds increase, losses mount with rising transmission



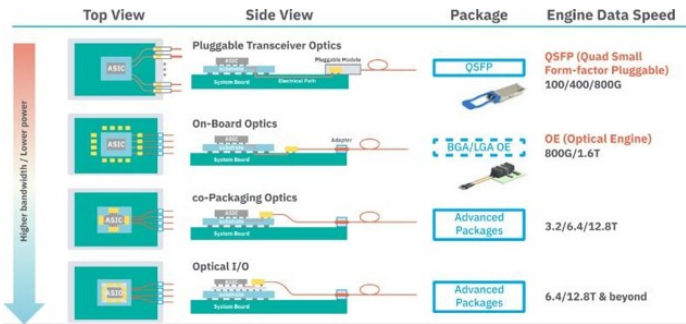
CPO is the most power saving solution



Migration from transceivers to CPO architecture



CPO evolution progress



Source: Broadcom, ASE, Morgan Stanley Research

Allring Benefited from CPO, New US Customers, and Ongoing CoWoS Capacity Expansion from Non-TSMC

Allring Revenue Bottom-up Assumption

(NT\$m)	2024	2025	2026e	2027e	2028e	2025 Y/Y	2026 Y/Y	2027 Y/Y	2028 Y/Y
CoWoS	3,530	4,771	8,055	13,500	14,400	35%	69%	68%	7%
CPO	0	150	200	3,000	4,500	NA	33%	1400%	50%
Flip Chip	200	0	900	200	0	-100%	NA	-78%	-100%
SolC		120	150	600	700		25%	300%	17%
US customer				200	200				
Others	1,804	325	750	500	800	-82%	131%	-33%	60%
Total Revenue	5,535	5,366	10,055	18,000	20,600	-3%	87%	79%	14%

Revenue mix	2024	2025e	2026e	2027e	2028e
CoWoS	64%	89%	80%	75%	70%
CPO	0%	3%	2%	17%	22%
Flip Chip	4%	0%	9%	1%	0%
SolC	0%	2%	1%	3%	3%
US customer	0%	0%	0%	1%	1%
Others	33%	6%	7%	3%	4%
Total Revenue	100%	100%	100%	100%	100%

Source: Company data, Yole, Morgan Stanley Research (e) estimates

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(as of September 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Charlie Chan		
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$154.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$4,345.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$744.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb981.00
CXMT Corp (688825.SS)	O (08/26/2026)	Rmb50.52
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT\$8,360.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$1,130.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$571.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$95.60
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$92.25
MediaTek (2454.TW)	O (11/28/2025)	NT\$4,690.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb467.58
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$532.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb78.96
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,060.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb96.00
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$462.50
SMIC (0981.HK)	O (10/21/2025)	HK\$56.80
TSMC (2330.TW)	O (02/07/2022)	NT\$2,550.00
UMC (2303.TW)	O (05/19/2026)	NT\$147.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$186.50
WIN Semiconductors Corp (3105.TWO)	E (10/09/2026)	NT\$581.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK\$163.80
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb52.02

Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb85.45
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb29.64
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb220.53
Innoscence (2577.HK)	E (10/13/2025)	HK\$37.20
JCET Group Co Ltd (600584.SS)	O (09/16/2026)	Rmb62.60
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$24.16
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb102.50
SJ Semiconductor Corp (688820.SS)	E (09/16/2026)	Rmb105.95
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb78.85
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb56.14
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb23.07
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb81.00
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$980.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,490.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$17,485.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$100.50
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$132.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT\$2,915.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb117.65
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb334.40
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$116.00
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$273.40
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb197.01
Novatek (3034.TW)	U (02/04/2026)	NT\$531.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$155.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$558.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$72.20
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$749.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb51.55
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$177.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$117.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$215.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb105.73
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb288.60
Henry Zhao		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$70.80
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb313.24

Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb64.65
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb600.32
Lucas Wang		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,525.00
FOCI Fiber Optic Communications Inc (3363.TWO)	E (10/09/2026)	NT\$667.00
Hon Precision (7769.TW)	O (04/17/2026)	NT\$5,925.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$304.50
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,575.00
Tiffany Yeh		
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$13.26
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$261.89
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$5,615.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.